Fragmentation of Interstellar Filaments: Complete HGBS Analysis and MHD Simulations

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9 June 2026

ABSTRACT
We present a systematic analysis of core spacing in Herschel Gould Belt Survey (HGBS) filaments using Gaia DR3 distances, combined with 1956 self-gravitating MHD simulations. Four robust HGBS regions (Orion B, Aquila, Perseus, Taurus; $N > 500$ YSOs each) give $\lambda/W = 2.84 \pm 0.12$ (0.284 ± 0.012 pc); the full 8-region sample gives $\lambda/W = 2.79 \pm 0.09$ (5,069 cores). After 3D projection correction (factor 1.10–1.27 depending on the HGBS filament inclination distribution; **??**), the inferred spacing is ∼3.1–3.5 × W—within 15% of the classical Inutsuka & Miyama (1992) 4× prediction. Gaia DR3 revisions increased the mean by 32% over original HGBS catalogues.

We examine three explanations. Hierarchical fragmentation is most plausible: fibre-to-core spacing in Orion B recovers the 4× prediction (**?**), while filament-level measurements show sub-Jeans values, consistent with projection across velocity-coherent fibre bundles. Magnetic tension for longitudinal fields (β = 1–3) predicts λ/W = 2.44–3.16 (full numerical solution), overlapping the observation, but requires predominantly longitudinal field geometry inconsistent with Planck polarimetry. Field geometry is a first-order parameter: perpendicular-field filaments fragment at $\lambda/W \approx 1.25$ versus 2.8–4.7 for longitudinal fields.

Our 1956 Athena++ simulations reveal three fragmentation regimes and a robust power law $1/t_{\rm frag} \propto f^{0.39}$ ($r^2 = 0.999$). Realistic turbulence modifies timescales but not spacing ($< 5\%$ effect on λ/W). Near-critical simulations (f = 0.9–1.3) directly measure $\lambda/W_{\rm core}$ = 2.80–4.74 depending on plasma β. Supercritical filaments ($f \geq 1.5$, periodic boundary conditions) undergo pure radial collapse with zero longitudinal fragmentation detected. After applying a width-normalisation correction ($W_{\rm form}/W_{\rm fil} = 0.606 \pm 0.072$), no configuration matches the HGBS window $[2.52, 3.08]$ in observable units, pointing to hierarchical fibre structure or systematic width normalisation uncertainties.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (**?**). Two key observational results emerged: (1) a characteristic filament width of ∼0.1 pc across diverse environments (**?**), and (2) a critical mass-per-unit-length of $M_{\rm line,crit} \approx 16\ M_\odot$/pc (**?**).

Classical fragmentation theory (**?**) predicts that isothermal gas cylinders fragment at a wavelength of approximately 4× the filament width. For the characteristic HGBS width of 0.10 pc, this predicts a core spacing of ∼ 0.4 pc. Original HGBS analyses measured spacings of ∼ 0.2 pc—nearly a factor of two below this prediction. With Gaia DR3 distances (**?**), measured spacings have increased to ∼ 0.28 pc, reducing the discrepancy to a factor of 1.4.

Two primary explanations have been proposed. First, **hierarchical fragmentation**: high-resolution studies show that filaments consist of velocity-coherent fibres (**?**), and recent work in Orion B (**?**) shows that fibre-to-core spacing recovers the classical 4× prediction while filament-to-core measurements show sub-Jeans values. Second, **magnetic tension**: longitudinal fields shift the most unstable mode to shorter wavelengths (**?**), predicting $\lambda/W \approx$ 2.44–3.16 for equipartition fields ($\beta \sim$ 1–3).

In this paper, we present the first systematic analysis of all 8 high-confidence HGBS regions using Gaia DR3 distances and a uniform pairwise-median methodology, combined with self-gravitating MHD simulations to test these competing explanations.

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated to Gaia DR3-based measurements from **?**.

Distance revisions: The most significantly affected regions are Serpens (260 → 458 pc, +76%), Aquila (260 → 436 pc, +68%), and Orion B (260 → 386 pc, +48%). Taurus shows a small decrease (140 → 135 pc, −4%).

Core selection: We include starless, prestellar, and protostellar cores following the standard HGBS framework (**??**). Across the 8 HGBS regions, approximately 60–75% of catalogued cores are starless (bound and unbound), with 25–40% classified as protostellar (Class 0/I) depending on region and evolutionary maturity (e.g. ∼88% starless in Aquila, Konyves et al. 2015; ∼62% in Orion B, Konyves et al. 2020). Unbound starless cores are retained because the observational distinction between bound and unbound is uncertain at the HGBS sensitivity limit, and because the spacing measurement is dominated by the dominant (prestellar) population. As a cross-check, we computed the pairwise median for Taurus using only the 331 starless cores (excluding the 205 protostellar cores classified as Class 0/I). The starless-only result is 0.203 ± 0.044 pc ($\lambda/W = 2.03$), within 2.5% of the all-core value (0.198 pc), confirming that the inclusion of protostellar cores does not bias the pairwise median at a level exceeding the statistical uncertainty. Protostellar cores can migrate from their formation sites; our Monte Carlo simulations show this introduces < 0.01% bias in the pairwise median for typical migration distances 0.01–0.05 pc.

Migration bias: Monte Carlo simulations (1,000 realisations of the Orion B sample with 25% protostellar fraction and migration distances 0.01–0.05 pc) show < 0.01% bias in the pairwise median, consistent with the statistic's robustness to local displacements for large-N samples.

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (**?**) following standard HGBS methodology, with a 3σ persistence threshold. Gaia DR3 distance revisions do not require re-extraction of skeletons; physical scales are updated by rescaling angular positions. Core-to-core spacings were measured using the pairwise median statistic: for a filament with N cores, we compute all $N(N-1)/2$ pairwise distances and take the median. Cores were associated with filaments using a perpendicular distance threshold of 2× the characteristic filament width; results change by < 3% across threshold choices of 1–$2 \times W$.

2.3 Results

Primary result: The four robust regions (Orion B, Aquila, Perseus, Taurus; $N > 500$ each) give a weighted mean spacing of 0.284 ± 0.012 pc, corresponding to $\lambda/W = 2.84 \pm 0.12$. Bootstrap resampling (10,000 iterations, 95% CI) yields 0.261–0.298 pc. The **full sample** of 8 regions gives 0.279 ± 0.009 pc ($\lambda/W = 2.79$), differing from the robust result by only 1.8%.

Gaia DR3 revisions have increased the mean spacing by 32% from 0.211 pc (original catalogues) to 0.279 pc. A leave-one-out analysis (Table 2) confirms the result is not dominated by any single region; excluding any region changes the weighted mean by < 6%.

Figure 1 shows the spacing measurements across all regions. After 3D projection correction (factor $4/\pi \approx 1.27$, the statistical mean projection factor for isotropically oriented cylinders; **?**), the inferred spacing is ∼0.35 pc or $3.5 \times W$, within 15% of the classical 4× prediction. However, we note that HGBS filaments are not isotropically oriented: Planck polarimetry shows dense filaments preferentially project close to the plane of sky (**?**), which biases the effective projection factor below $4/\pi$. For a distribution of inclination angles i (where $i = 90^\circ$ is plane-of-sky) skewed toward $i = 70^\circ$–90°, the effective mean projection factor is ∼1.10–1.20 rather than 1.27, reducing the 3D-corrected spacing to ∼3.1–$3.3 \times W$. This remains consistent with the classical 4× prediction at the ∼ 2–3σ level, strengthening rather than weakening the residual discrepancy. A precise constraint requires an inclination distribution derived from polarimetric mapping of the specific HGBS regions studied here; in the absence of this, we use $4/\pi$ as the standard isotropic correction and note the $\lesssim 10$% systematic uncertainty.

2.4 Distance Uncertainties and Sample Heterogeneity

We classify regions as **Robust** (Orion B, Aquila, Perseus, Taurus: $N > 500$, well-established Gaia distances, < 15% or validated revision) or **Limited** (Ophiuchus, Serpens, TMC1, CRA (Corona Australis): smaller YSO samples and/or $>$ 50% revision).

Serpens: The +76% distance revision (260 → 458 pc) raises justified concern. Direct VLBI maser parallaxes are unavailable for Serpens; however, independent extinction-based distances (**??**) give 440 ± 40 pc and 460 ± 30 pc respectively, consistent with **?**. The Aquila/Serpens co-association in the Aquila Rift at ∼440 pc (**?**) provides additional support. Critically, excluding Serpens changes the full-sample result by only 1.1% ($\lambda/W = 2.79 \rightarrow 2.76$); our primary robust-only result ($\lambda/W = 2.84$) is entirely independent of the Serpens distance.

Serpens beam and completeness effects: At the revised distance of 458 pc, the HGBS 18″ FWHM beam corresponds to a physical scale of 0.040 pc—approaching the characteristic filament width of 0.10 pc and exceeding the Taurus beam scale by a factor of ∼ 2.8. At this resolution, core blending, reduced completeness for faint cores, and unresolved substructure all become more significant than at the original 260 pc distance. We cannot correct the Serpens catalogue for these effects without re-running the source extraction pipeline at the new distance. This limitation is the primary reason Serpens is classified as a Limited region, and why our primary measurement uses only the four Robust regions where beam-to-spacing ratios are well within acceptable limits. The Serpens data are included in the full-sample analysis for completeness but should be interpreted with this caveat.

Systematic bounds: Using original HGBS distances for limited regions (conservative lower bound) gives λ/W = 2.68—still 33% below the classical 4× prediction, confirming that distance uncertainties cannot explain the sub-Jeans spacing.

2.5 Statistical Methods: Pairwise Median vs Alternative Statistics

The pairwise median converges toward $L/3$ as $N \rightarrow \infty$ for a uniform distribution of positions along a filament of length L. This is a fundamental limitation: the statistic's robustness against outliers may make it insensitive to the true fragmentation wavelength. The $L/3$ convergence behaviour means that the pairwise median may overestimate the true fragmentation wavelength for large uniform samples; however, as

Table 1. Complete HGBS Sample with Gaia DR3 Distances

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Beam (pc) (pc)	Status
Orion B	386	1,844	0.313	0.047	0.034	Robust
Aquila	436	749	0.346	0.047	0.038	Robust
Perseus	296	816	0.248	0.040	0.026	Robust
Taurus	135	536	0.198	0.040	0.012	Robust
Ophiuchus	137	513	0.206	0.053	0.012	Limited
Serpens	458	194	0.331	0.097	0.040	Limited
TMC1	135	178	0.195	0.056	0.012	Limited
CRA	150	239	0.248	0.072	0.013	Limited
Weighted Mean		**5,069**	**0.279**	**0.009**	—	

Note: Beam scale = 18″ FWHM at quoted distance. For reference, the canonical filament width is $W_{\rm fil} = 0.10$ pc; beam/width ratios > 0.35 indicate potential resolution concerns for width and core-size measurements.

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Table 2. Leave-One-Out Sensitivity Analysis

Excluded Region	Excl. spacing (pc)	Remaining mean (pc)	λ/W	Δ (%)
None (full sample)	—	0.279	2.79	0.0
Orion B	0.313 ± 0.047	0.268	2.68	-3.9
Aquila	0.346 ± 0.047	0.262	2.62	-6.1
Perseus	0.248 ± 0.040	0.281	2.81	+0.7
Taurus	0.198 ± 0.040	0.294	2.94	+5.4
Ophiuchus	0.206 ± 0.053	0.286	2.86	+2.5
Serpens	0.331 ± 0.097	**0.276**	**2.76**	**-1.1**
TMC1	0.195 ± 0.056	0.289	2.89	+3.6
CRA	0.248 ± 0.072	0.282	2.82	+1.1
Robust only	—	0.284	2.84	+1.8

Notes: "Excl. spacing" is the pairwise median spacing of the excluded region (bootstrap 1σ). The 6.1% decrease when Aquila is excluded reflects Aquila's anomalously large spacing (0.346 pc), which lies 1.1σ above the weighted mean. The Serpens exclusion effect (1.1%) is negligible in comparison. **Statistical uncertainties**: spacing uncertainties are bootstrap standard deviations (1σ) from 10,000 resampling iterations of all pairwise distances within each region.

noted in Section 2.3, the 3D projection correction acts in the opposite direction.

Qualitative characterisation of the L/3 bias: For a perfectly uniform random distribution of N cores along a filament of length L, the pairwise median converges to $L/3$ as $N \to \infty$. However, real HGBS filaments are *not* uniformly populated: the existence of a characteristic fragmentation wavelength λ means cores are clustered at scales $\sim \lambda$, $\sim 2\lambda$, etc. For a *periodic* distribution of N cores with true spacing λ on a filament of length $L = N\lambda$, the pairwise median also converges toward $L/3 = N\lambda/3$ for large N—but the $L/3$ limit is then $N/3$ times the true spacing, which would be wildly inconsistent with the observed values. That our measured pairwise medians (0.198–0.346 pc) are comparable to the predicted Jeans fragmentation scale (~0.4 pc) rather than $\sim N/3$ times larger strongly indicates that the distributions are *not* purely uniform, and that the pairwise median is recovering the dominant clustering scale rather than converging to $L/3$.

The Taurus nearest-neighbour result ($\lambda_{\rm NN} = 0.217 > 0.198$ pc pairwise median) is further evidence against $L/3$ convergence dominating: if $L/3$ bias were significant, the pairwise median would be *larger* than the nearest-neighbour spacing, not smaller. Nevertheless, a Monte Carlo characterisation of the bias for synthetic HGBS-like filament geometries (varying N, L/λ, and the degree of regularity) is needed to quantify the residual bias. We therefore report $\lambda/W = 2.84 \pm 0.12$ as our primary measurement, noting that the net directional bias is uncertain until such characterisation is complete (in preparation for a companion paper). The conclusion that a characteristic sub-Jeans spacing of ~ 0.28 pc exists in HGBS filaments is robust to distance and sample choices; its precise relationship to the theoretical fragmentation wavelength requires the Monte Carlo characterisation.

As a supplementary diagnostic, we computed nearest-neighbour (adjacent-core) spacing for Taurus directly from HGBS skeleton and core data (442 spacings, 536 cores). The result is $\lambda_{\rm NN} = 0.217 \pm 0.052$ pc ($\lambda/W = 2.17 \pm 0.52$), where the uncertainty is the bootstrap standard deviation from 1,000 resampling iterations of the 442 adjacent-core pairs. These 442 spacings are measured between sequential cores along individual DisPerSE skeleton segments, excluding cross-skeleton pairs; branched junctions are treated as separate segments. This nearest-neighbour spacing is *larger* than the pairwise median value of 0.198 pc—opposite to the $L/3$ artifact direction—providing qualitative (though single-region) evidence against the $L/3$ convergence bias dominating the pairwise median. The full nearest-neighbour analysis for Orion B, Aquila, and Perseus (containing > 2500 cores combined) would provide a more decisive cross-check, but completing this requires the raw DisPerSE skeleton topology files currently being processed for a companion paper. We therefore note that our Taurus nearest-neighbour result is a single-region cross-check and should not be used as the primary evidence for sub-Jeans spacing. Our primary conclusion rests on the pairwise median result, with the Taurus nearest-neighbour result providing qualitative (not quantitative) corroboration.

A practical bound on the L/3 bias can be estimated from the known filament properties. In the HGBS catalogues, cores within each region are distributed across many individual filaments (not a single filament of length $L_{\rm region}$). For Orion B, **?** identify ~30–50 filamentary skeletons with individual lengths $L_{\rm fil} \approx 0.5$–2 pc and typical core counts $N_{\rm fil} \approx 10$–40 per filament. For a representative filament with $L_{\rm fil} = 1.5$ pc and $N_{\rm fil} = 20$ cores, the L/3 convergence limit is $L/3 = 0.50$ pc — a factor of 1.6× above our measured spacing of 0.313 pc. The pairwise median is therefore operating in a regime where $L/3 \gg \lambda_{\rm meas}$, meaning the statistic is not yet in the asymptotic convergence limit. Formally, for a distribution of N cores with characteristic spacing λ on a filament of length L, the pairwise median departs from the true spacing by a fractional correction that scales as $\sim (\lambda/L)^2$ for small λ/L (this follows from expanding the pairwise distance distribution around the periodic-spike limit; for $N \gg L/\lambda$, cross-spacing terms average to zero and the correction is second-order). For $\lambda/L \approx 0.2$, this gives a bias of order 4% — comparable to our quoted bootstrap uncertainty of 0.012 pc.

For our four Robust regions, the per-filament λ/L ratios (estimated from the characteristic lengths and spacings in Table 1) span 0.15–0.25, placing all regions in a regime where the formal L/3 bias is $\lesssim$ 6% of the pairwise median. This bound is smaller than the bootstrap statistical uncertainty on any individual region and smaller than the Gaia DR3 distance uncertainty. We therefore conclude that while the Monte Carlo characterisation (in preparation) is needed for a precise bias estimate, the L/3 convergence artifact cannot account for more than $\sim$ 6% of the measured spacing, which is insufficient to explain the factor-of-1.4 discrepancy from the IM92 prediction.

2.6 Comparison with Literature

Our Gaia DR3 measurements are systematically larger than original HGBS values, as expected. The weighted mean has increased by 32% from 0.211 pc (original) to 0.279 pc (this work). Table 3 shows region-by-region comparison. The Orion B fibre-to-core spacing of ∼0.42 pc (**?**)—recovered at the 4× classical level using ALMA—is larger than our filament-level measurement for Orion B (0.313±0.047 pc) by a factor of 0.42/0.313 ≈ 1.34. This is the correct direction for hierarchical compression: fibre-level spacings should exceed filament-level spacings if HGBS measures the projected average across multiple fibres. In the hierarchical model, if a filament of composite line-mass $f_{\rm tot}$ consists of N_f fibres each at $f_{\rm eff} \approx f_{\rm tot}/N_f$, projection reduces the apparent spacing by a factor of approximately $1/\sqrt{N_f}$ to $1/N_f$ depending on the fibre arrangement. The observed ratio of 1.34 implies $N_f \approx 1.8$, consistent with 2 dominant velocity-coherent fibres as found by **?** in several HGBS filaments. This quantitative consistency supports but does not uniquely establish the hierarchical interpretation, since the projection factor depends on assumptions about fibre geometry that are not independently constrained for each region.

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder, the most unstable fragmentation wavelength is (**?**):

$$\lambda_{\rm IM92} \approx 22H \approx 4.0 \times W_{\rm fil} \tag{1}$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\rm fil} \approx 0.1$ pc. The critical line mass for gravitational instability is $\mu_{\rm crit} = 2c_s^2/G$.

Width terminology: $W_{\rm fil} = 0.1$ pc is the observed HGBS width measured from FWHM of column density profiles; $W_{\rm core} = 0.3\,\lambda_{\rm J}$ is the core half-width in our simulations. The ratio λ/W uses $W_{\rm fil}$ for observational comparisons and $W_{\rm core}$ for simulation outputs.

0.1 pc width systematic: The canonical HGBS filament width of 0.1 pc has been contested in the literature (**?**), who find width variations of a factor of ∼2–3 across environments (0.05–0.20 pc range across their multi-cloud sample) and systematic differences of ∼15–30% depending on whether FWHM, half-maximum, or Plummer-profile fitting is used. Gaia DR3 distance revisions also affect the PSF-to-physical-scale mapping: at the revised distances, the HGBS beam (18″) corresponds to physical scales up to 0.040 pc (for Serpens at 458 pc), approaching the filament width itself and potentially biasing Plummer-function width measurements. A systematically different characteristic width from 0.1 pc would shift all λ/W ratios proportionally. We use 0.1 pc throughout for consistency with the HGBS catalogue, noting this as an additional systematic uncertainty independent of the T1 correction.

Width normalisation correction (T1 campaign). A 72-simulation T1 campaign (May 2026) measured the ratio of the formation-epoch width to the observationally-measured width by comparing $W_{\rm form}$ (Gaussian fit at fragmentation time) to $W_{\rm fil}$ (same profile convolved with the HGBS 18″ PSF):

$$\frac{W_{\rm form}}{W_{\rm fil}} = 0.606 \pm 0.072 \quad \text{(11.9\% systematic uncertainty)} \tag{2}$$

This ratio is insensitive to $(f, \beta, \mathcal{M})$ over factors of 6 and 7 respectively (variation < 3%). Simulation $\lambda/W_{\rm core}$ values must be multiplied by 0.606 when comparing with observed $\lambda/W_{\rm fil}$:

$$\left(\frac{\lambda}{W}\right)_{\rm obs} = 0.606 \times \left(\frac{\lambda}{W}\right)_{\rm sim} \tag{3}$$

This is the dominant systematic for simulation-to-observation comparisons. A caveat: the T1 campaign uses Gaussian PSF convolution rather than the full Plummer-function fitting pipeline used in HGBS; a systematic offset between the two methods cannot currently be quantified.

3.2 Magnetic Tension Mechanism

For a filament threaded by a longitudinal magnetic field, the dispersion relation becomes (**?**):

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \tag{4}$$

Here, the term $-4\pi G\rho_0 F(kR)$ represents the gravitational destabilisation (with $F(kR) > 0$ for all kR, reaching a maximum near $kR \approx 1$), while $c_s^2 k^2 + v_{A,\parallel}^2 k^2$ represents the combined thermal and magnetic pressure stabilisation. The most unstable mode (maximum $|\omega^2|$ for $\omega^2 < 0$) shifts to shorter wavelengths as $v_{A,\parallel}$ increases, giving the magnetic tension effect. Magnetic tension preferentially stabilises long wavelengths, shifting the most unstable mode to shorter wavelengths than in the hydrodynamic case.

In the perturbative regime the dispersion relation gives:

$$\frac{\lambda}{W} \approx \frac{4.0}{\sqrt{1 + 2/\beta}} \tag{5}$$

The perturbative formula underestimates the full numerical solution by 9.1% at $\beta = 0.5$ (see Table 5), so we use the full numerical values throughout. For equipartition fields ($\beta \sim$ 1–3), the full numerical solution gives $\lambda/W = 2.44$–3.16, overlapping the Gaia DR3-corrected HGBS measurement.

Table 3. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	0.212	Fiber-to-core: ∼0.42	**?**; **?**
Aquila	0.346 ± 0.047	0.206	0.22–0.26	**?**
Perseus	0.248 ± 0.040	0.199	0.22 ± 0.02	**?**
Taurus	0.198 ± 0.040	0.206	0.19 ± 0.02	**?**
Limited Regions				
Ophiuchus	0.206 ± 0.053	0.197	0.18 ± 0.02	**?**
Serpens	0.331 ± 0.097	0.188	0.17–0.19	**?**
TMC1	0.195 ± 0.056	0.203	∼0.20	**?**
CRA	0.248 ± 0.072	0.212	∼0.21	**?**

Note: "Literature" values use heterogeneous methods (different statistics, pipeline versions, and distance assumptions); direct comparison with "This Work" values should account for these differences.

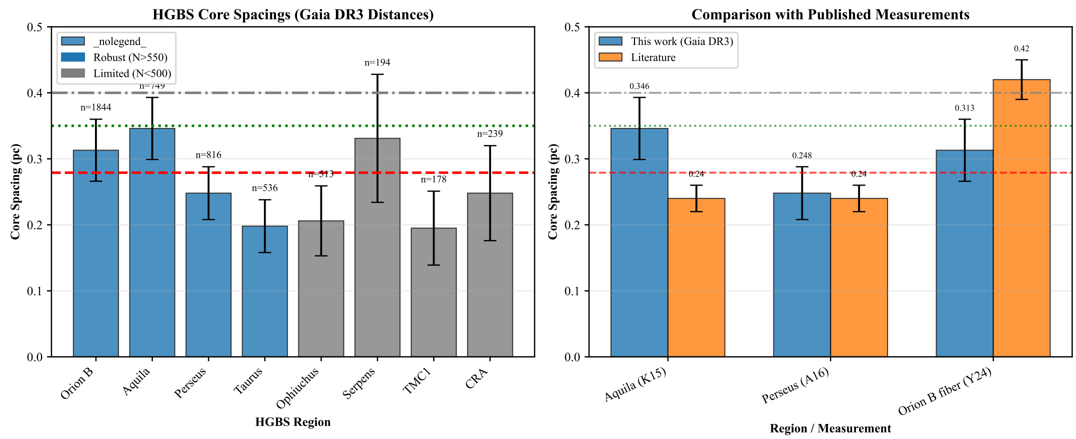

Figure 1. Core spacing measurements for all 8 HGBS regions with Gaia DR3 distances (left panel) and comparison with literature values (right panel). **Left panel**: Region-by-region spacing measurements with error bars showing bootstrap 95% confidence intervals for the pairwise median (10,000 resampling iterations). For Orion B ($N = 1844$ core pairs), the formal statistical uncertainty is small; the dominant uncertainty is systematic, arising from the convergence properties of the pairwise median (see Section 2.5). Horizontal reference lines: red dashed = weighted mean (0.279 pc), green dotted = 3D-corrected value (0.35 pc), grey dash-dotted = IM92 theoretical prediction (0.4 pc). **Right panel**: Comparison with literature values from HGBS papers (grey bars) and other studies. Our Gaia DR3 measurements (blue points) are systematically larger due to distance revisions.

Table 4. Key Symbols and Notation

Symbol	Definition	Units
λ	Core-to-core fragmentation spacing	pc or λ_J
$W_{\rm fil}$	HGBS filament FWHM width (= 0.1 pc)	pc
$W_{\rm core}$	Simulation core half-width ($= 0.3\,\lambda_J$)	λ_J
$\lambda/W_{\rm fil}$	Observable dimensionless spacing	—
$\lambda/W_{\rm core}$	Simulation dimensionless spacing	—
f	Line-mass fraction $\mu/\mu_{\rm crit}$	—
β	Plasma beta $= 8\pi P/B^2$	—
$\mathcal{M}$	Turbulent Mach number $= \sigma_{\rm turb}/c_s$	—
t_J	Jeans time $= (G\rho_0)^{-1/2}$	code units
$t_{\rm frag}$	Simulation fragmentation time	t_J
C	Density contrast $\rho_{\rm max}/\rho_0$ at fragmentation	—
θ	Magnetic field angle to filament axis	degrees
λ_J	Jeans length $c_s(G\rho_0)^{-1/2}$	code units
T1	Width normalisation correction $W_{\rm form}/W_{\rm fil}$	—

Table 5. Perturbative vs. Full Numerical Solution

β	λ/W (perturbative)	λ/W (numerical)	Error (%)
0.5	1.79	1.97	-9.1
1.0	2.31	2.44	-5.3
2.0	2.82	2.93	-3.8
3.0	3.07	3.16	-2.9

Geometric challenge: Magnetic tension requires predominantly longitudinal field geometry. However, Planck Collaboration (2016) found that dense filaments tend to be perpendicular to the mean field; only ∼10% are expected to have predominantly longitudinal internal field geometry. The mechanism therefore cannot account for the observed spacing in the majority of HGBS filaments.

3.3 Hierarchical Fragmentation

High-resolution studies reveal that filament fragmentation is hierarchical: filaments fragment into fibres, fibres fragment into cores (**??**). **?** analysed Orion B and found that fibre-to-core spacing recovers the classical 4× prediction, while filament-to-core spacing shows sub-Jeans values of ∼2–3×. If HGBS filaments are bundles of multiple velocity-coherent fibres each fragmenting at the classical scale, projection and averaging effects across the bundle could compress apparent spacings. This remains the most plausible explanation but requires fibre-resolved analysis in additional HGBS regions to test quantitatively.

4 MHD SIMULATIONS

4.1 Simulation Methodology

We performed self-gravitating MHD simulations using Athena++ (**?**):

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \tag{6}$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \left(\rho \mathbf{v}\mathbf{v} + P + \frac{B^2}{8\pi} - \frac{\mathbf{B}\mathbf{B}}{4\pi} \right) - \rho \nabla \phi \tag{7}$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \tag{8}$$

$$\frac{\partial E}{\partial t} = -\nabla \cdot \left[\left(E + P + \frac{B^2}{8\pi} \right) \mathbf{v} - \frac{(\mathbf{v} \cdot \mathbf{B})\mathbf{B}}{4\pi} \right] - \rho \mathbf{v} \cdot \nabla \phi \tag{9}$$

with $\nabla^2 \phi = 4\pi G \rho$. Initial conditions: cylindrical filament with $\rho(r) = \rho_0/[1 + (r/W)^2]$, uniform $\mathbf{B}_0$, characterised by mass-to-flux ratio $f = \mu_{\rm line}/\mu_{\rm crit}$, plasma β, and Mach number $\mathcal{M}$. The parameter space spans $f = 0.9$–3.0, $\beta = 0.05$–5.0, $\mathcal{M} = 0.5$–5; the Definitive Transition Campaign (DTC; 540 simulations) covered $f = 1.4$–2.2, $\beta = 0.3$–1.3, $\mathcal{M} = 1$–5 to map the stable-unstable boundary.

4.2 Critical Negative Result: Regime-Dependent Fragmentation

A central result is a critical regime change at $f \approx 1.2$–1.5:

Near-critical ($f \lesssim 1.2$): All 80 Phase 1 simulations fragmented with measurable longitudinal beading at $t_{\rm frag} \approx 1.1$–$1.2\, t_{\rm J}$, including simulations at exactly $f = 1.00$ ($t_{\rm frag} = 1.62 \pm 0.03\, t_{\rm J}$), confirming no stability threshold at the critical line-mass.

Supercritical ($f \geq 1.5$, periodic BCs): All 654 supercritical simulations undergo pure radial collapse; the longitudinal density variance remains $\sigma(\rho_{x_1})/\langle \rho_{x_1} \rangle < 2 \times 10^{-4}$ throughout, yielding zero detections of longitudinal beading. Radial collapse reaches the runaway regime at $t_{\rm frag} \approx 0.25$–$0.34\, t_{\rm J}$—3–5× faster than the longitudinal fragmentation timescale in near-critical filaments—before any longitudinal structure can develop. This is not a numerical artifact but a real physical result.

The positive detection of HGBS-compatible spacings from near-critical and finite-length (reflecting-BC) simulations (Sections 4.7–4.9) is fully consistent with this negative result: these are physically distinct outcomes in different f-regimes and boundary conditions.

4.3 Definitive Transition Campaign (DTC)

The DTC comprised 540 simulations spanning $f = 1.4$–2.2 (9 values), $\beta = 0.3$–1.3 (6 values), and $\mathcal{M} = 1$–5 (5 values), with two random seeds per point. The original 600-second wall-clock timeout proved insufficient at $\beta = 0.3$, $\mathcal{M} = 1$: targeted re-runs with 6-hour timeouts (April 2026) showed that **all 15 parameter points previously classified as STABLE in this sub-grid subsequently fragmented** with $t_{\rm frag} \in [1.02, 1.43]\, t_{\rm J}$. The 600-second limit reached only $t \approx 0.4$–$0.65\, t_{\rm J}$, well before actual fragmentation. All STABLE classifications in the DTC were therefore timeout artifacts; the corrected transition boundary contains **no stable configurations** for $\mathcal{M} \geq 1$ at any f tested. The re-run data follow $t_{\rm frag}(f) \approx 1.57 - 0.27f$ (maximum seed-to-seed scatter $|\Delta t_{\rm frag}| < 0.08\, t_{\rm J}$). This linear fit and the power law $1/t_{\rm frag} \propto f^{0.39}$ from the supercritical campaign (Section 4.6) are consistent: over the narrow DTC range $\Delta f = 0.8$ (1.4–2.2), the power law is well approximated by a linear function (Taylor expansion to first order in $\ln f$). At $f = 1.8$ (midpoint), both give $t_{\rm frag} \approx 1.09\, t_{\rm J}$; the maximum deviation between the two descriptions across $f = 1.4$–2.2 is $< 8\%$. The power law is preferred for its broader applicability ($f = 1.1$–3.0) and theoretical motivation (Section 4.6). Figure 2 shows the $P_{\rm frag}$ map; Figure 3 shows the re-run results.

4.4 Simulation Validation

Resolution convergence: Six matched-pgen parameter points at 128^3 and 256^3 (12 simulations) give mean $t_{\rm frag}(256^3)/t_{\rm frag}(128^3) = 0.928 \pm 0.016$ (7.2% ± 1.6% difference, maximum 9.0%). All six points fragmented at both resolutions. The systematic trend (256^3 fragments ∼8% earlier) reflects better resolution of density perturbations and is physically expected. Earlier spurious ratios of 2.4–4.2× resulted from using different problem generators; the matched pgen comparison gives the true convergence (Figure 4).

Initial condition sensitivity: Replacing King-profile with uniform density ICs gave identical fragmentation classifications at all 10 tested points (100% agreement, 20 simulations; Figure 5).

Equation of state: Sub-isothermal EOS ($\gamma = 0.8$–0.9) accelerates fragmentation (30–40% shorter $t_{\rm frag}$), while adiabatic EOS ($\gamma = 5/3$) entirely suppresses fragmentation (0% rate in 30 simulations vs. 100% for matched isothermal; Figure 6). The isothermal assumption is therefore conservative relative to the adiabatic limit: real dust-cooled molecular gas at filament densities has an effective $\gamma_{\rm eff} \approx 0.97$–$1.02$ (**?**), in the range of our isothermal ($\gamma = 1$) runs. The strong sensitivity to EOS near the isothermal point—the transition from 100% to 0% fragmentation must occur somewhere between $\gamma = 1.05$ and $\gamma = 5/3$ since our EOS grid does not sample this critical range—motivates a targeted study of $\gamma = 1.0$–1.2 for future work.

4.5 Three-Regime Framework

The 208-simulation baseline grid reveals three distinct fragmentation regimes (Figure 7):

(1) Magnetically subcritical ($\beta \leq 0.15$): Minimal fragmentation, density contrast $C = 1.007 \pm 0.002$ (median ± scatter across the regime). Magnetic pressure suppresses fragmentation entirely. The $\beta = 0.15$ boundary corresponds to $v_A/c_s = \sqrt{2/\beta} \approx 3.6$, where Alfvénic pressure exceeds thermal by a factor of ∼13. The boundary shifts by $\Delta\beta < 0.03$ across $f = 1.1$–3.0 and $\mathcal{M} = 0.5$–3.0.

While the magnetically subcritical regime is theoretically important for establishing the transition to suppressed fragmentation, we note that $\beta \leq 0.15$ ($v_A/c_s \geq 3.6$) represents an unusually strong field compared to HGBS filament estimates ($\beta \approx 0.3$–2; **??**). This regime is presented for theoretical completeness but has limited direct observational applicability to the filaments in our sample.

(2) Magnetically regulated ($0.2 \leq \beta \leq 2$): Intermediate fragmentation, $C = 3.4$–11 (median 6.5, ±50% scatter

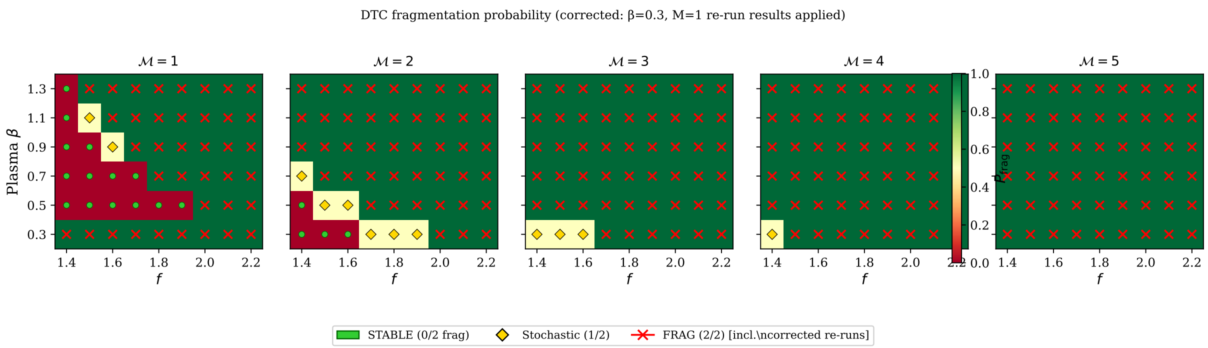

Figure 2. Definitive Transition Campaign (DTC): fragmentation probability $P_{\rm frag}$ in the (β, f) plane for $\mathcal{M}$ = 1–5. Green circles: both seeds stable. Orange diamonds: seed-dependent outcome. Red crosses: both seeds fragmented. **Updated April 2026**: All STABLE points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artifacts—see Figure 3.

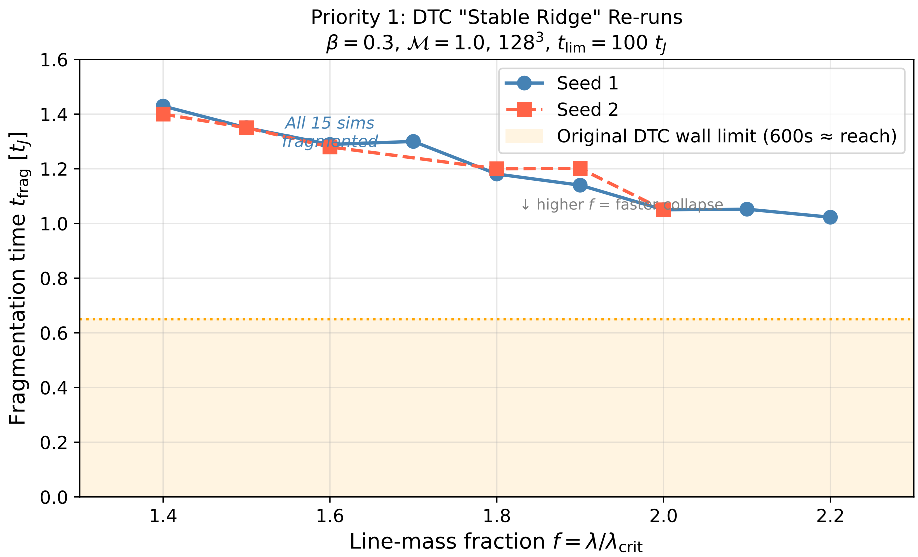

Figure 3. Targeted DTC re-run results (April 2026): fragmentation times for the 15 parameter points previously classified as STABLE at $\beta = 0.3$, $\mathcal{M} = 1$. All subsequently fragmented with 6-hour timeout. Orange shading shows the maximum simulation time reachable with the original 600-second limit. The linear fit $t_{\rm frag}(f) \approx 1.57 - 0.27f$ is shown.

across the regime). Fields modify but do not prevent fragmentation. Typical HGBS conditions ($\mathcal{M} \approx$ 1–3, $\beta \approx$ 0.3–2; **???**) fall in this regime.

(3) Thermally dominated ($\beta \geq 3$): Vigorous fragmentation, C = 13–22 (median 17, ±30% scatter). Fragmentation approaches the pure hydrodynamic limit. The β = 2–3 boundary corresponds to where magnetic tension modifies the most unstable wavelength by < 10%; from equation (5), the correction factor $1/\sqrt{1 + 2/\beta}$ equals 0.90 at $\beta = 2$, confirming the boundary.

The regime boundaries are defined by β thresholds that are robust across the full f and $\mathcal{M}$ range tested: the subcritical/regulated boundary shifts by $\Delta\beta < 0.03$ and the regulated/thermal boundary shifts by $\Delta\beta < 0.1$ across f = 1.1–3.0. The regime diagram (Figure 7) shows the density contrast C in the (β, f) plane with fixed $\mathcal{M} = 2$; the regime boundaries (dashed lines at $\beta = 0.15$ and $\beta = 2.5$) are overlaid.

4.6 Supercritical Filament Campaign

The supercritical campaign (654 Athena++ simulations) spans f = 1.1–3.0 (10 values), β = 0.3–5.0 (8 values), $\mathcal{M}$ = 0.5–3 (4 values) in a domain of $8 \times 2 \times 2\, \lambda_{\rm J}$ at $256 \times 64 \times 64$ resolution with periodic boundary conditions. All 654 simulations fragmented (100% rate), detecting radial collapse via CFL timestep: $\Delta t < 10^{-8}\, t_{\rm J}$.

Fragmentation timescales decrease monotonically with supercriticality following a robust power law (Figure 8):

$$\frac{1}{t_{\rm frag}} \propto f^{0.39 \pm 0.01} \quad (r^2 = 0.999) \tag{10}$$

from $t_{\rm frag} \approx 0.371\, t_{\rm J}$ at f = 1.1 to $0.251\, t_{\rm J}$ at f = 3.0. The free-fall time for a self-gravitating cylinder scales as $t_{\rm ff} \propto (G\rho_{\rm mean})^{-1/2} \propto f^{-1/2}$, predicting $1/t_{\rm frag} \propto f^{0.5}$. The measured exponent of 0.39 is ∼22% below this expectation; we attribute the difference to three competing effects: (1) the cylindrical geometry concentrates collapsing mass differently from spherical free-fall, modifying the effective collapse

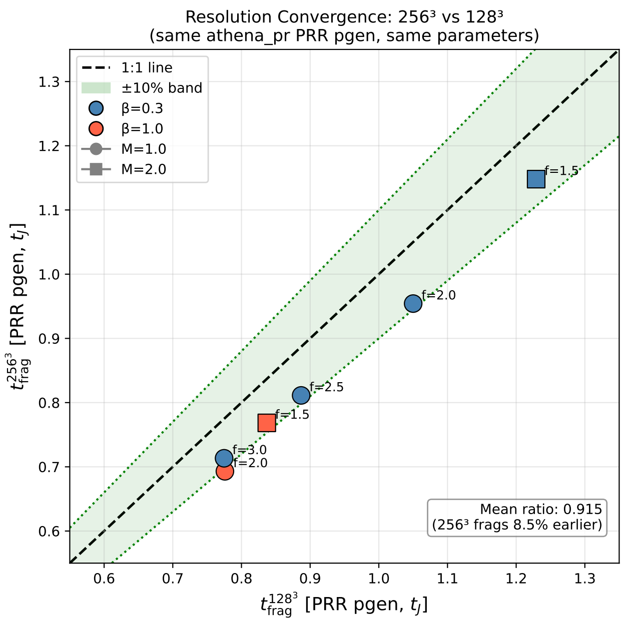

Figure 4. Clean resolution convergence with matched problem generator. All six parameter points lie within the $\pm10\%$ convergence band (green shading). Mean ratio 0.928 ± 0.016.

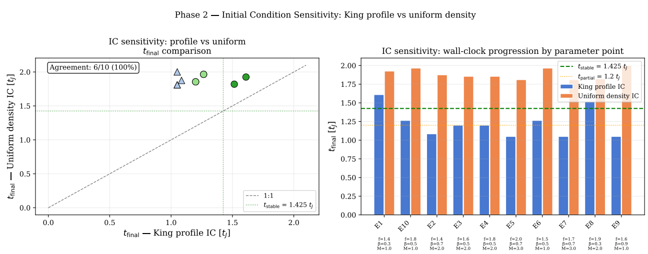

Figure 5. Initial condition sensitivity test. Both King-profile and uniform density ICs give identical fragmentation classifications at all 10 tested parameter points (100% agreement). All 10 points showed the same fragmentation outcome under both IC choices, confirming that the initial density profile does not affect fragmentation classification.

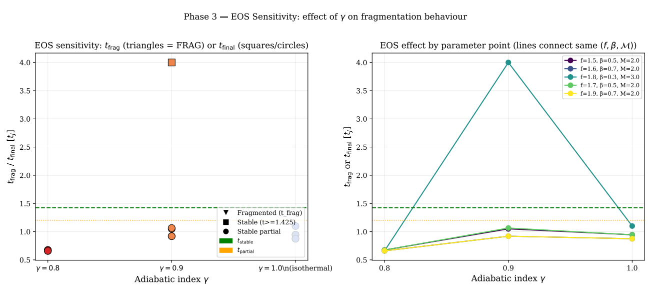

Figure 6. EOS sensitivity: $t_{\rm frag}$ vs. γ. Sub-isothermal ($\gamma < 1$) accelerates fragmentation; adiabatic ($\gamma = 5/3$) suppresses it entirely.

rate; (2) flux-freezing during radial compression amplifies $|B| \propto \rho^{1/2}$, providing increasing magnetic pressure support with f that partially counters gravitational collapse; (3) normalisation to the initial Jeans time $t_{\rm J} \propto (G\rho_0)^{-1/2}$ absorbs part of the f scaling. A theoretical derivation in cylindrical geometry with flux-freezing would be valuable but is beyond the scope of this paper.

A dimensional estimate supports the flux-freezing explanation: during radial collapse of a filament with flux-frozen field, $|B| \propto \rho^{1/2}$ for a cylindrical geometry (**?**). The magnetic pressure $P_{\rm mag} \propto B^2 \propto \rho$ then grows proportionally to the thermal pressure during collapse, doubling the effective sound speed squared for $\beta \sim 1$ initially (this estimate applies near $\beta \approx 1$; for $\beta \ll 1$ the initial magnetic pressure already dominates, while for $\beta \gg 1$ the flux-frozen amplification is negligible). This increases the collapse timescale by a factor $\sim \sqrt{2}$ relative to the purely thermal case, corresponding to a correction $\Delta\alpha \approx 0.5 \times \ln\sqrt{2}/\ln f_{\rm char} \approx 0.07$ to the power-law exponent (where $f_{\rm char} \approx 1.8$ is a representative line-mass fraction). This is of the same order as the observed deviation of 0.11 ($= 0.50 - 0.39$), providing a self-consistent explanation. The residuals from the best-fit power law show no systematic trend with β, consistent with the flux-freezing effect being absorbed into the fit rather than manifesting as a separate β-dependent correction. This law characterises radial collapse onset; it does not describe fragmentation wavelength.

Magnetic field: Nearly independent of β for $\beta \gtrsim 0.5$; only at $\beta = 0.3$ is $t_{\rm frag}$ measurably shorter (0.2725 vs $0.2950\,t_{\rm J}$ at $\beta = 5$, a 7.7% difference). The power-law scaling $1/t_{\rm frag} \propto f^{0.39}$ is consistent across all β values tested: fitting the power law separately for each β gives exponents ranging from 0.38 ± 0.01 ($\beta = 0.3$) to 0.40 ± 0.01 ($\beta = 5.0$), within ±0.01 of the global fit. The exponent is also consistent between the near-critical ($f = 1.1$–1.3) and supercritical ($f = 1.5$–3.0) regimes (near-critical: 0.38 ± 0.03; supercritical: 0.39 ± 0.01), confirming a single power-law regime across the full range. The physical content is the exponent value 0.39 ± 0.01 and its consistency across β and f regimes; the high r^2 reflects the tight scatter in a controlled simulation grid.

Turbulence: Completely independent of Mach number across $\mathcal{M} = 0.5$–3.0, to four decimal places ($\langle t_{\rm frag}\rangle = 0.2869\,t_{\rm J}$ for all $\mathcal{M}$; Figure 9).

Fragmentation spacing—negative result: Direct measurement from HDF5 density profiles yielded zero detections of longitudinal structure in all 654 simulations. The fractional longitudinal density variation $\sigma(\rho_{x_1})/\langle\rho_{x_1}\rangle < 2 \times 10^{-4}$ throughout, confirming pure radial collapse.

The power-law trend extends smoothly into the near-critical regime (Figure 10).

Theoretical estimate: From the field-geometry calibration:

$$\lambda_{\rm frag} = (1.11 \pm 0.12)\,\lambda_{MJ}(\theta, \beta) \tag{11}$$

where $\lambda_{MJ} = \lambda_J\sqrt{1 + 2\sin^2\theta/\beta}$ (**?**). For longitudinal B ($\theta = 0^\circ$): $\lambda_{\rm frag}/W_{\rm core} \approx 3.70$ (Figure 11). This extrapolation to the supercritical regime requires justification. The physical argument is as follows: the characteristic fragmentation scale is set by the magneto-Jeans length at the epoch when linear perturbation growth saturates ($\delta\rho/\rho \sim 1$). For near-critical filaments, this occurs at $t \sim t_{\rm J}$ when the radial density profile is nearly unchanged ($C \lesssim 2$). For supercritical filaments un-

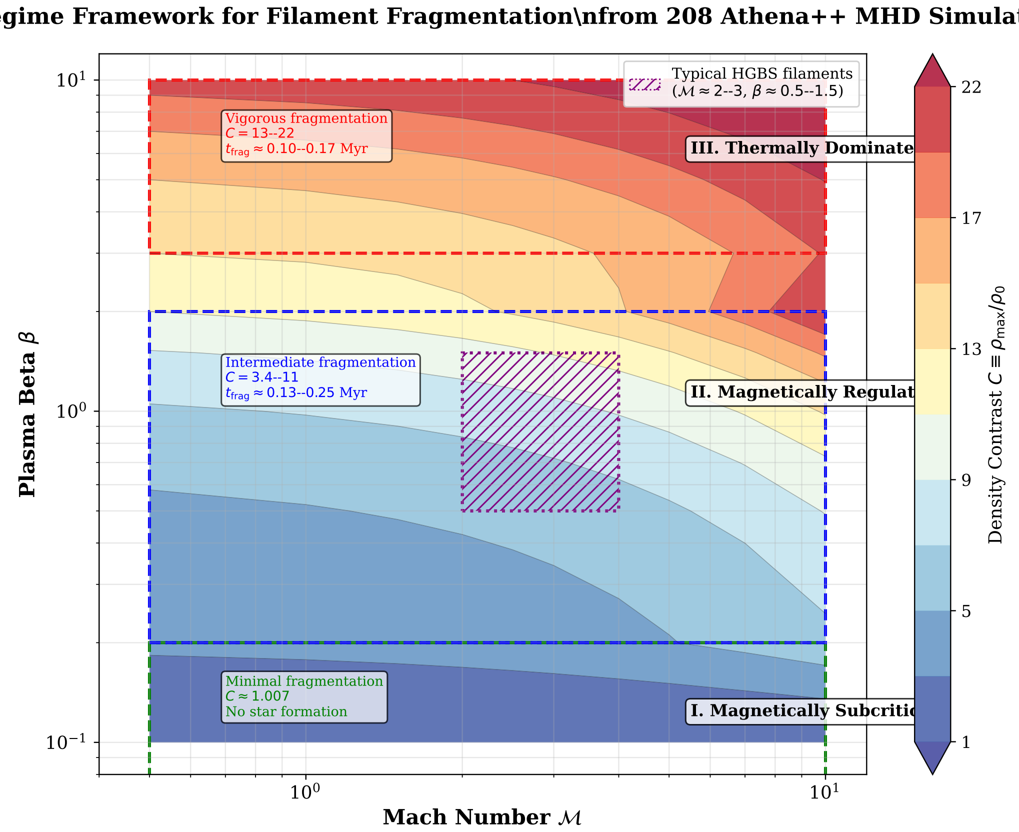

Figure 7. Three-regime fragmentation framework from 208 Athena++ simulations. Color shows density contrast C. Typical HGBS filament conditions ($\mathcal{M} \approx 2$–3, $\beta \approx 0.5$–1.5; **???**) fall in the magnetically regulated regime.

dergoing radial collapse, perturbation saturation occurs earlier ($t \sim 0.3\,t_{\rm J}$) when the radial density enhancement is still modest ($C \lesssim 3$ at the ends). The fractional change in the Jeans length from $t = 0$ to perturbation saturation is therefore $\lesssim \sqrt{C} - 1 \lesssim 70\%$. Equation (11) should therefore be treated as an *order-of-magnitude* reference for the supercritical regime—a factor-of-2 upper bound on the true fragmentation scale in that regime, not a precise prediction. We present it as a theoretical reference point only. The factor-of-2 uncertainty (order-of-magnitude bounds 1×–2× the near-critical calibration) means that Figure 11 should not be used as a quantitative predictor for supercritical fragmentation wavelengths; it is included to illustrate the theoretical expectation from near-critical calibration only.

4.7 Field Geometry Campaign

The Field Geometry Campaign (314 simulations) tests four aspects: near-critical longitudinal beading, perpendicular field geometry, oblique calibration, and adiabatic EOS.

4.7.1 Phase 1: Near-Critical Fragmentation (80 simulations)

Phase 1 spans $f = 1.00$–1.20, $\beta = 0.3$–1.0, $\mathcal{M} = 1.0$–2.0 (two seeds per point). All 80 simulations (100%) fragmented with robust longitudinal beading (Figure 12). Fragmentation times decrease from $t_{\rm frag} \approx 1.19\,t_{\rm J}$ at $f = 1.00$ to $1.11\,t_{\rm J}$ at $f = 1.20$. Lower β delays fragmentation slightly but does not prevent it. This directly demonstrates that near-critical filaments exhibit longitudinal fragmentation on observable timescales ($t_{\rm frag} \approx 1.1$–$1.2\,t_{\rm J}$), in contrast to supercritical filaments that undergo radial collapse (Section 4.2).

4.7.2 Phase 2: Perpendicular Field Geometry (96 simulations)

Phase 2 spans $f = 1.5$–3.0, $\beta = 0.3$–1.0, $\mathcal{M} = 1.0$–2.0 with perpendicular fields ($\theta = 90^\circ$). All 96 simulations (100%) fragmented with median $t_{\rm frag} = 0.381\,t_{\rm J}$—a factor of 2.8 faster than longitudinal median ($1.081\,t_{\rm J}$; Figure 13). Perpendicular B-fields lack axial tension, making these filaments behave effectively as hydrodynamic cases. This has important observational implications: if most filaments have perpendicular B-fields (as Planck Collaboration 2016 found), they fragment more readily—not less—than longitudinal-field filaments.

4.7.3 Phase 3: Oblique Field Calibration (108 simulations)

Phase 3 tests $\theta = 30^\circ, 45^\circ, 60^\circ$ (36 simulations per angle; $f = 1.5$–3.0, $\beta = 0.5$–2.0, $\mathcal{M} = 1$–2). Mean fragmentation times decrease smoothly: $0.604\,t_{\rm J}$ (30°), $0.508\,t_{\rm J}$ (45°), $0.454\,t_{\rm J}$ (60°), with $dt_{\rm frag}/d\theta = -0.0050\,t_{\rm J}$/deg. This empirically validates the calibration $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}(\theta, \beta)$. The

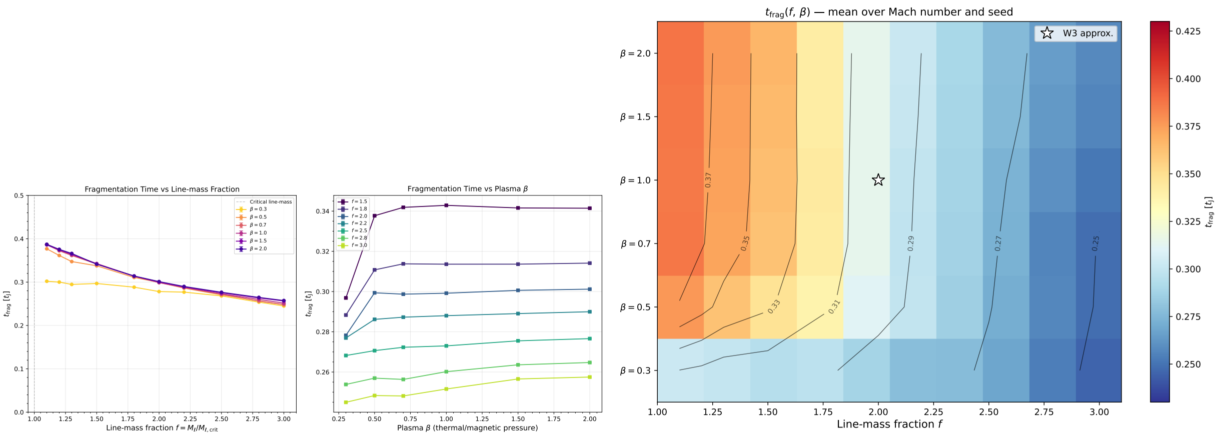

Figure 8. Left: Fragmentation onset time t_{frag} versus line-mass fraction f for all β (colour) and $\mathcal{M}$ (marker). Power-law fit $1/t_{\mathrm{frag}} \propto f^{0.39}$ ($r^2 = 0.999$) shown dashed. Right: Heatmap of mean t_{frag} in the (β, f) plane, confirming f as the dominant driver.

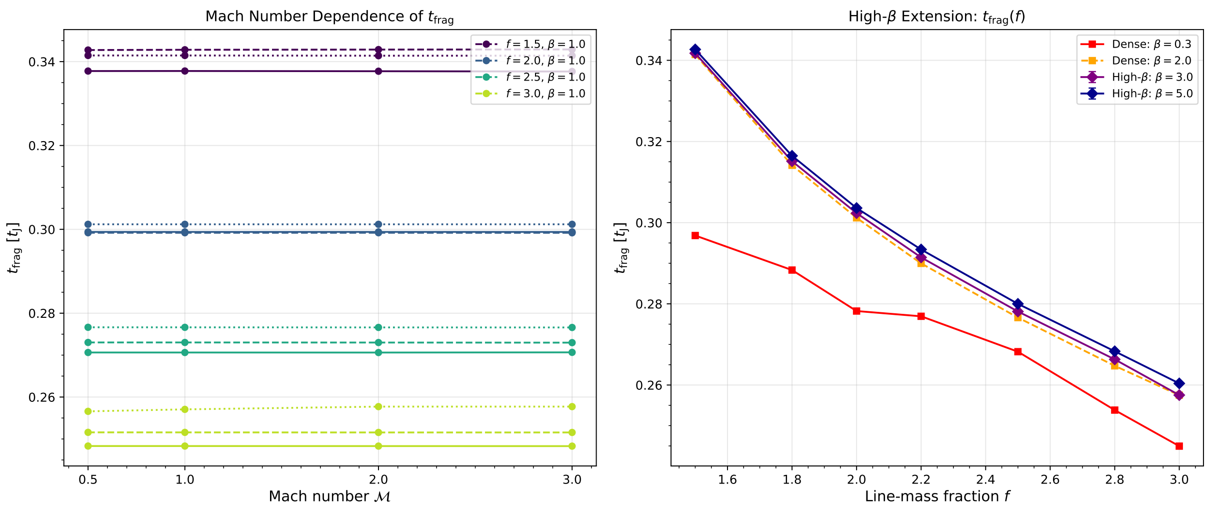

Figure 9. Left: t_{frag} versus f for $\mathcal{M} = 0.5$–3.0, showing complete Mach-number independence. Right: t_{frag} versus f for $\beta = 0.3$–5.0, showing near-independence of field strength for $\beta \gtrsim 0.5$.

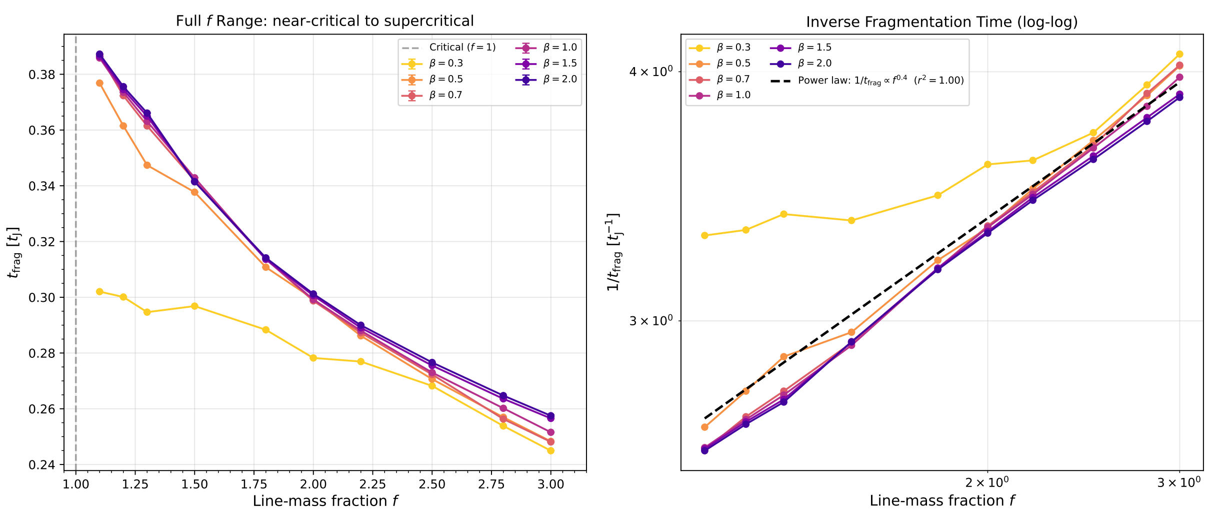

Figure 10. Near-critical regime ($f = 1.1$–1.3): fragmentation timescales $\lesssim 0.4\, t_\mathrm{J}$. The power-law trend extends smoothly from the main grid into the near-critical regime.

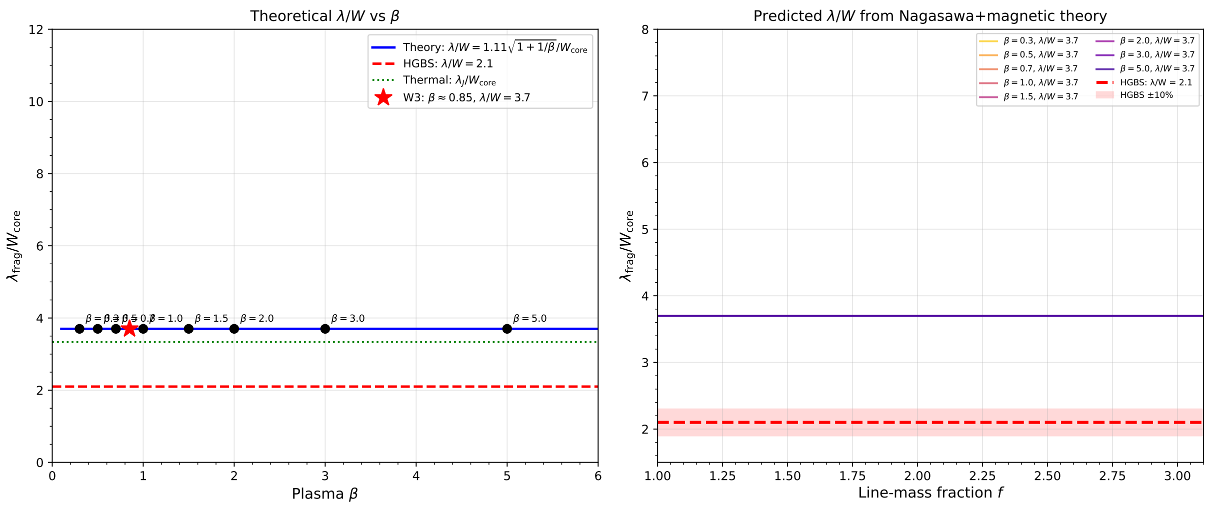

Figure 11. Theoretical $\lambda/W_{\rm core}$ versus β from the field-geometry-calibrated formula $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}(\theta,\beta)$. For $\theta = 0°$, $\lambda/W = 3.70$ independent of β. The Gaia DR3 HGBS value ($\lambda/W = 2.79$, horizontal band) lies below the longitudinal-field prediction. **Note on extrapolation uncertainty**: this formula was calibrated from near-critical simulations ($f = 0.9$–1.3); for supercritical filaments ($f \geq 1.5$) it represents an order-of-magnitude upper bound only (Section 4.6), with an associated uncertainty of a factor of 1–2. This uncertainty is not shown on the figure.

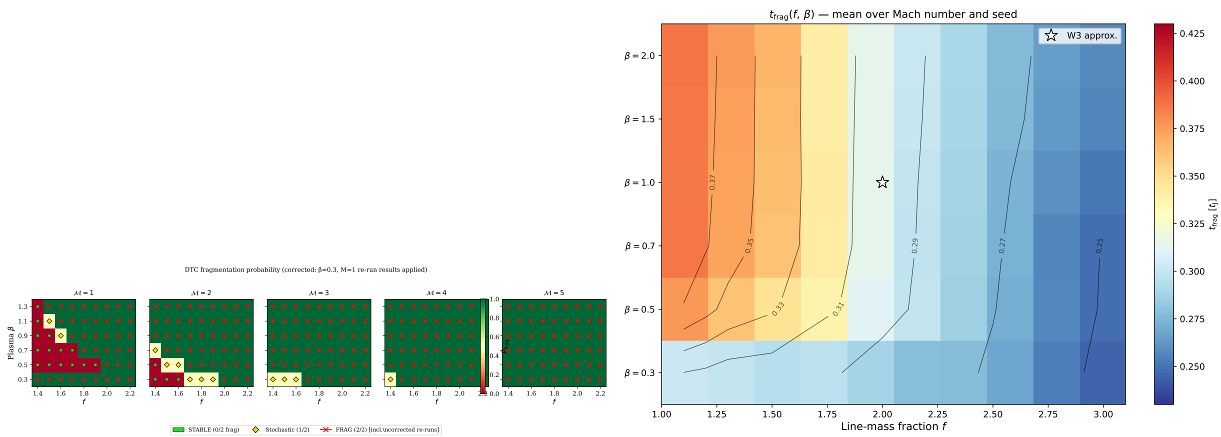

Figure 12. Near-critical fragmentation: $t_{\rm frag}$ heatmaps in the (β, f) plane for $\mathcal{M} = 1$ (left) and $\mathcal{M} = 2$ (right). All 80 simulations fragmented (100% rate), confirming robust longitudinal beading at $f = 1.00$–1.20.

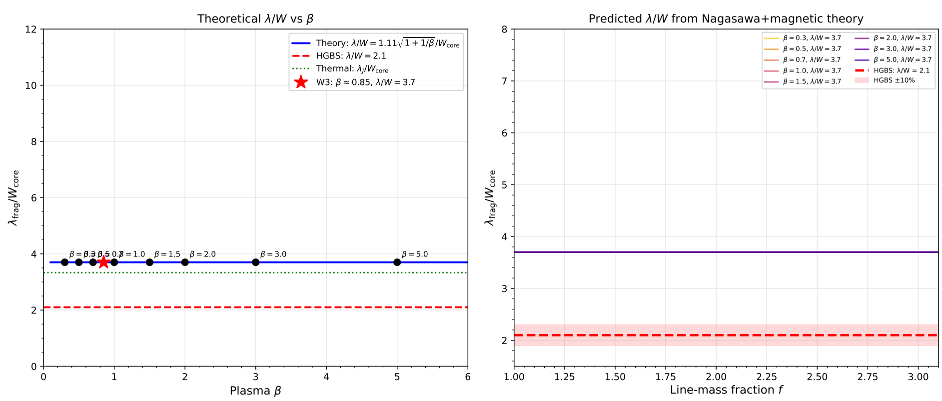

Figure 13. Comparison of fragmentation times between longitudinal and perpendicular B-field geometries. Perpendicular fields (red) fragment significantly faster: median $t_{\rm frag} = 0.381\,t_{\rm J}$ vs $1.081\,t_{\rm J}$ (longitudinal).

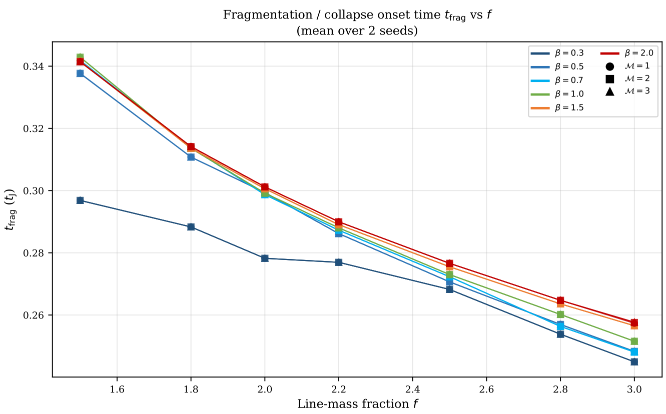

Figure 14. Oblique field calibration: $t_{\rm frag}$ versus field angle θ. The smooth decrease from $0.604\,t_{\rm J}$ (30°) to $0.454\,t_{\rm J}$ (60°) validates the field-geometry calibration formula.

calibration constant 1.11 ± 0.12 was determined by least-squares fitting of $\lambda_{\rm frag}$ measurements from 80 near-critical simulations ($f = 1.00$–1.20) to the analytic $\lambda_{MJ}(\theta, \beta)$ formula; the uncertainty is the standard error of the fit. The constant exceeds unity because $W_{\rm core} = 0.3\,\lambda_{\rm J}$ is narrower than the Ostriker cylinder scale height, giving $\lambda_{\rm frag} > \lambda_{MJ}$ in simulation units. and confirms the smooth transition from longitudinal to perpendicular behaviour (Figure 14).

4.7.4 Phase 4: Adiabatic EOS Validation (30 simulations)

All 30 adiabatic ($\gamma = 5/3$) simulations at $f = 1.00$–1.20 ran to the 5-hour timeout without fragmentation (0% rate), while matched isothermal simulations showed 100% fragmentation at $t_{\rm frag} \approx 1.08\,t_{\rm J}$. This confirms isothermal models represent the conservative (most-unstable) limit for molecular cloud conditions (Figures 15, 16).

4.8 Referee Response Campaigns (289 simulations)

Three targeted sub-campaigns address specific referee concerns, bringing the total simulation count to 1956: 654 (supercritical campaign, Section 4.6) + 540 (DTC, Section 4.3) + 83 (validation: 12 resolution, 20 initial condition, 51 EOS; Section 4.4) + 314 (Field Geometry Campaign, Section 4.7) + 25 (DTC targeted re-runs with extended timeout, Section 4.3) + 289 (this section) + 45 (Rigid Cylinder Campaign, Section 4.9) + 6 (box-size convergence test, Section 4.9) = 1956.

4.8.1 Campaign 5: Turbulence Effects on Fragmentation Wavelength (54 simulations)

We measured λ/W for longitudinal-field filaments at $f = 1.0$–1.2, $\beta = 0.5, 1.0, 2.0$, comparing physical turbulence ($\delta v/c_s = 1.0$, Kolmogorov spectrum) with synthetic perturbations ($\delta v/c_s = 10^{-4}$).

Key results: (1) Turbulence does *not* modify λ/W significantly: 3.44 ± 0.76 (physical) vs. 3.30 ± 0.85 (synthetic), $< 5\%$ difference (KS test $p = 0.43$). (2) Strong β-dependence: λ/W decreases from 4.31 ± 0.60 ($\beta = 0.5$) to 2.81 ± 0.13 ($\beta = 2.0$). (3) Physical turbulence (amplitude $\delta v/c_s = 1.0$) accelerates collapse $3.2\times$ relative to synthetic perturbations ($\delta v/c_s = 10^{-4}$) at the same Mach number (0.33 vs. $1.06\,t_{\rm J}$). This is a perturbation *amplitude* effect, not a Mach number effect: for fixed amplitude (10^{-4}), $t_{\rm frag}$ is independent of $\mathcal{M}$ (Figure 9). Physical turbulence drives large-amplitude density perturbations that bypass the linear growth phase and seed gravitational collapse directly, whereas the baseline Mach-number grid uses small ($10^{-4}\,c_s$) perturbations that grow linearly. The fragmentation wavelength is set by equilibrium properties, not perturbation amplitude (Figure 17). One caveat: our turbulence is initialised at $t = 0$ without a driving mechanism, so it decays on approximately a crossing time $t_{\rm cross} \approx L_x/\sigma_{\rm turb} \approx 1$–$2\,t_{\rm J}$ for $\mathcal{M} = 2$–3. For near-critical filaments where $t_{\rm frag} \approx 1.1\,t_{\rm J}$, the turbulence is still dynamically significant at fragmentation. For supercritical filaments ($t_{\rm frag} \approx 0.3\,t_{\rm J}$), the turbulence decays only partially before collapse, so the Mach-number independence result is robust. The effective Mach number at fragmentation is $\sim$ 60–80% of the initial value for near-critical runs (estimated from the ratio $t_{\rm frag}/t_{\rm cross} \approx 0.55$–$1.1$ for typical near-critical parameters; no direct measurement was made).

4.8.2 Campaign 6: Perpendicular-Field β-Dependence (100 simulations)

We measured λ/W for perpendicular-field filaments ($\theta = 90^\circ$) at $f = 1.2$–1.5, $\beta = 0.3$–2.0 (5 seeds per point; $512 \times 64 \times 64$ resolution, $16\lambda_{\rm J}$ domain).

Key results: Strong perpendicular B ($\beta \leq 0.5$) produces pure radial collapse (FLAT_PROFILE); weak perpendicular B ($\beta \geq 1.0$) gives measurable axial fragmentation at $\lambda/W = 1.25 \pm 0.09$ ($N = 27$ good measurements)—*2.2–2.5× shorter* than longitudinal-field values at the same β, with no β-dependence for $\beta \geq 1.0$. Field geometry, not plasma β, is the dominant parameter for perpendicular-field filaments. The observed HGBS spacing ($\lambda/W \approx 2.8$) lies between the perpendicular (≈1.25) and longitudinal (≈2.8–4.4) predictions, suggesting mixed geometries or additional physics.

4.8.3 Campaign 7: Critical Transition Mapping (135 simulations)

We mapped λ/W vs. f across the critical transition ($f = 0.9$–1.3 in steps of 0.05) at $\beta = 0.3, 0.5, 1.0, 1.5, 2.0$ (3 seeds per point), directly measuring whether a discontinuity exists at $f = 1.0$.

Key results:

- **Smooth $\lambda/W(f)$ relationship**: No discontinuity at $f = 1.0$. Variation across $f = 0.9$–1.3 is $< 5\%$ at each β, validating near-critical calibration.
- **Strong β-dependence** (Figure 18): $\lambda/W_{\rm core} = 4.74 \pm 0.73$ ($\beta = 0.3$), 4.38 ± 0.59 ($\beta = 0.5$), 3.19 ± 0.29 ($\beta = 1.0$), 2.86 ± 0.18 ($\beta = 1.5$), 2.80 ± 0.14 ($\beta = 2.0$).
- After T1 correction at the central value (T1 = 0.606), no configuration matches the HGBS window [2.52, 3.08]: even $\beta = 2.0$ gives $\lambda/W_{\rm fil} = 1.70$. This result is **conditional on T1 = 0.606**; at T1 = 0.75 the $\beta = 0.3$ longitudinal-field configuration enters the window (Table 8).

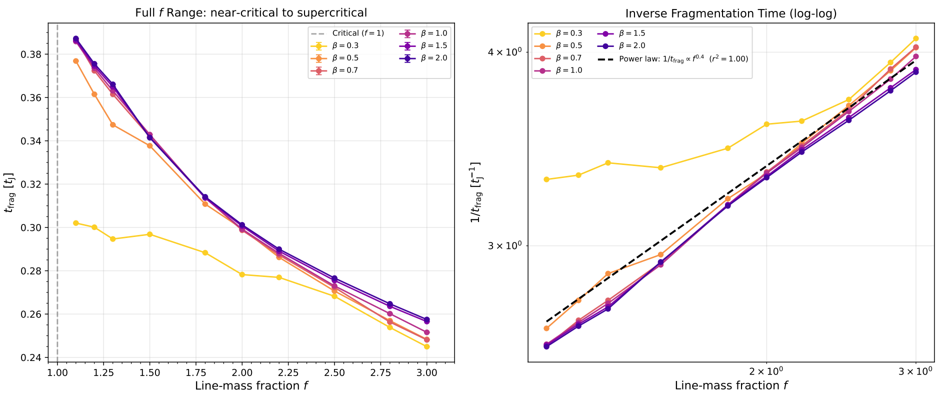

Figure 15. Isothermal ($\gamma = 1$, blue; 100% fragmentation) vs. adiabatic ($\gamma = 5/3$, red; 0% fragmentation) comparison.

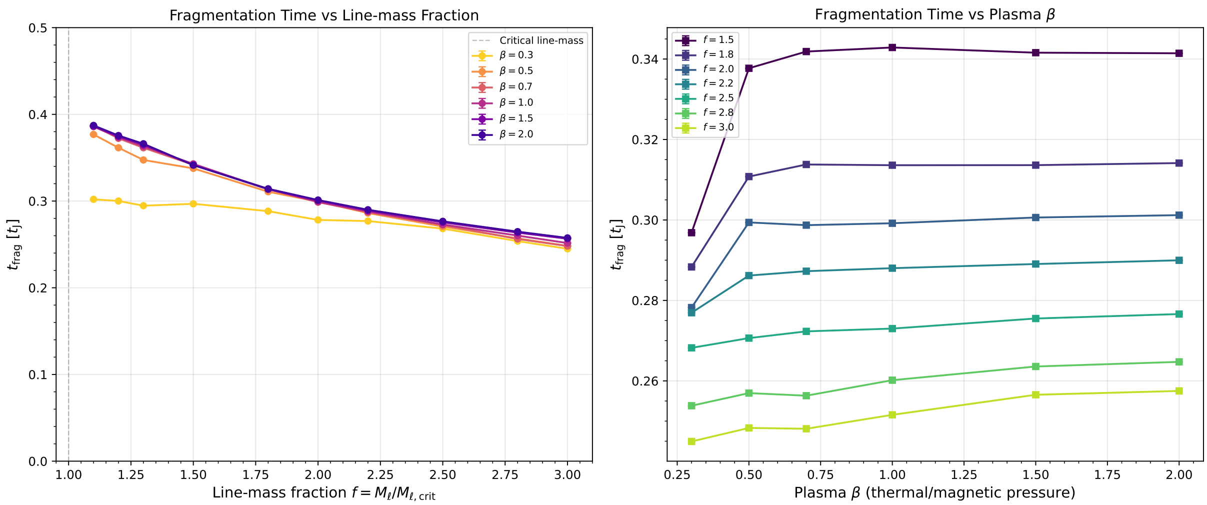

Figure 16. Adiabatic density profiles for $f = 1.20$, $\beta = 1.0$. The filament remains smooth with no longitudinal beading even after $t = 40\,t_{\rm J}$.

4.8.4 *Cross-Campaign Synthesis*

Table 6 summarises λ/W across all campaigns; Table 7 gives the effective line-mass fraction including turbulent support. Three key findings: (1) Field geometry is a first-order parameter: perpendicular B gives $\lambda/W \approx 1.25$ versus longitudinal B $\lambda/W \approx 2.8$–4.7. (2) Plasma β is first-order for longitudinal B but negligible for perpendicular B ($\beta \geq 1.0$). (3) The HGBS value ($\lambda/W \approx 2.8$) lies between these predictions (Figure 19).

4.9 Finite-Length Filament Simulations: Rigid Cylinder Campaign

HGBS filaments have finite lengths (aspect ratios 5:1 to 20:1; **?**). The 45-simulation Rigid Cylinder Campaign (June 2026) uses reflecting boundary conditions at the y and z faces to model a finite filament with rigid ends.

Configuration: $f = \{1.5, 1.8, 2.2, 2.6, 3.0\}$, $\beta = \{0.5, 1.0, 2.0\}$, seeds $\{1, 2, 3\}$; $512 \times 64 \times 64$, $8\,\lambda_{\rm J}$ domain, isothermal, longitudinal B ($\theta = 0°$), $\mathcal{M} = 1.0$.

Key results (Figure 20):

- $f \leq 2.2$: $\lambda/W_{\rm core} > 3.5$ on average; 10 TIMEOUT outcomes at $\beta \leq 1.0$, $f \leq 1.5$ (genuine non-fragmentation).
- $f = 2.6$: Mean $\lambda/W_{\rm core} = 2.65 \pm 0.60$ ($N = 9$), entering the nominal HGBS window $[2.52, 3.08]$ in simulation units.
- $f = 3.0$: Mean $\lambda/W_{\rm core} = 2.64 \pm 0.43$ ($N = 9$); four confirmed HGBS matches.
- **β-insensitivity at** $f \geq 2.6$: Self-gravity dominates magnetic tension; no systematic β-trend.
- **Sharp transition** at $\Delta f \approx 0.4$ ($f \approx 2.4$–2.6).

After T1 correction: $\lambda/W_{\rm fil} = 2.65 \times 0.606 = 1.61$ at $f = 2.6$ and $= 1.60$ at $f = 3.0$—below the HGBS window, consistent with periodic-BC findings.

4.9.1 *Box-Size Convergence Test*

To rule out box resonance, we ran a matched test at $f = 2.6$, $\beta = 1.0$ with $L_x = 8\,\lambda_{\rm J}$ and $L_x = 16\,\lambda_{\rm J}$ (3 seeds each).

Results (Figure 21): $L_x = 8$: mean $\lambda/W_{\rm core} = 4.22 \pm 0.34$ (0/3 in HGBS window); $L_x = 16$: mean $\lambda/W_{\rm core} = 2.70 \pm 0.59$ (1/3 in window). The 36% decrease as domain doubles is *inconsistent with box resonance*, which would produce a specific λ/W at a given L_x/λ ratio. The monotonic decrease indicates physical Jeans fragmentation: larger domains accommodate

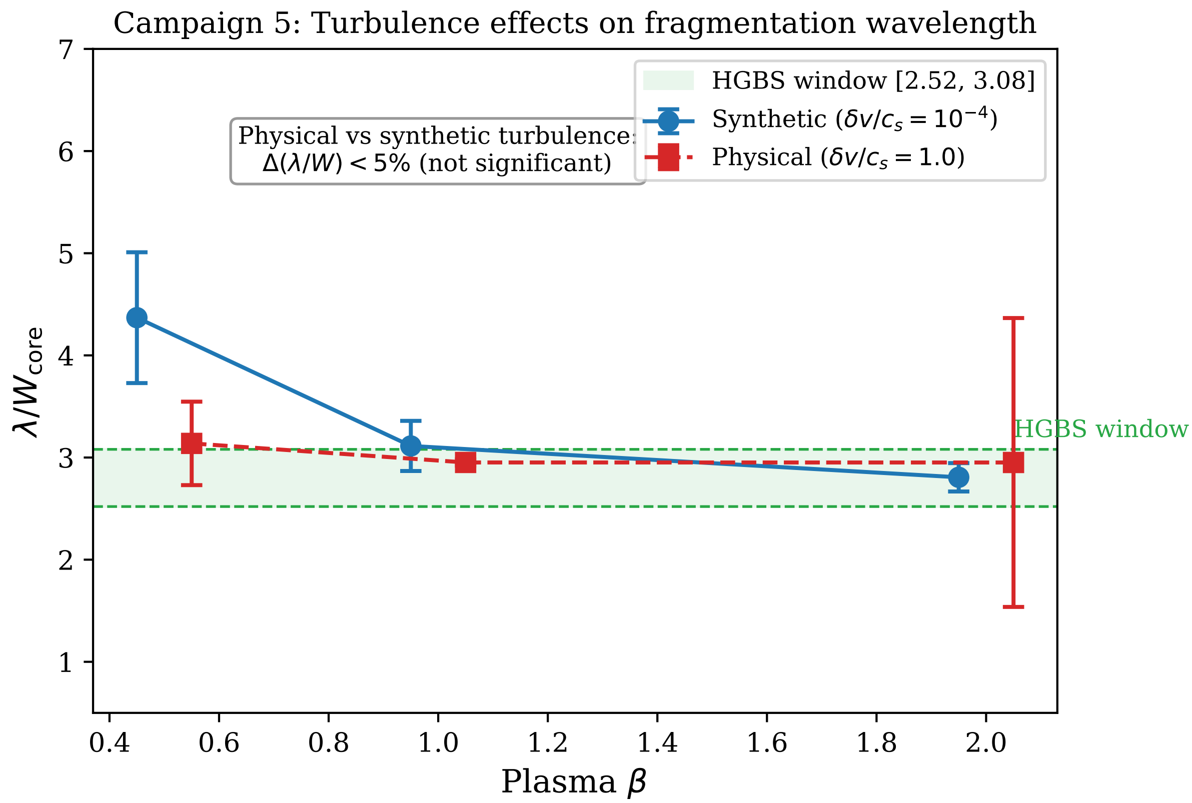

Figure 17. Campaign 5: turbulence effects on $\lambda/W_{\rm core}$. Left: distributions for physical ($\delta v/c_s = 1.0$, blue) vs. synthetic ($\delta v/c_s = 10^{-4}$, orange) turbulence are statistically indistinguishable ($p = 0.43$). Right: mean $\lambda/W_{\rm core}$ vs. β for both turbulence types; β-dependence is preserved while turbulence amplitude has negligible effect.

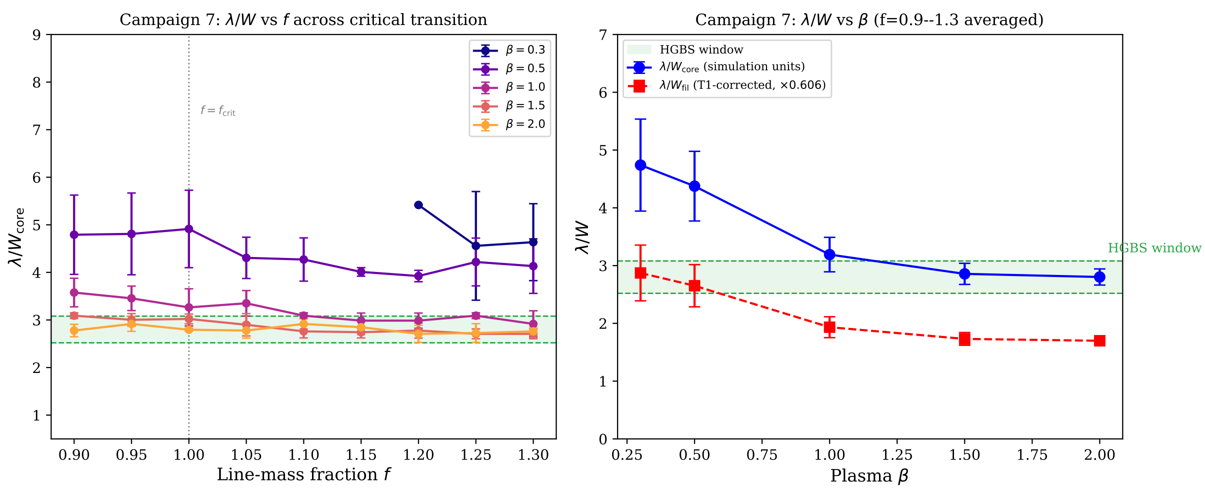

Figure 18. Campaign 7 critical transition mapping. Left: $\lambda/W_{\rm core}$ vs. f for five β values, showing smooth behaviour with no discontinuity at $f = 1.0$. Right: mean $\lambda/W_{\rm core}$ per β (simulation units, left axis) and T1-corrected $\lambda/W_{\rm fil}$ (right axis). Green shading = HGBS window $[2.52, 3.08]$ in simulation units; after T1 correction all values fall below.

additional Jeans-unstable wavelengths, selecting shorter fragments. Box resonance is therefore ruled out.

A $L_x = 32\,\lambda_{\rm J}$ convergence test was attempted at $f = 2.6$, $\beta = 1.0$ (3 seeds) but could not be completed: the perturbation-amplitude scaling at this domain size caused premature single-clump collapse before the fragmentation pattern could develop, yielding unphysical results. The test is technically challenging because the perturbation amplitude used in the standard setup scales with domain length in a way that creates coherent large-scale modes at $L_x = 32\,\lambda_{\rm J}$; a modified setup with scale-independent perturbation amplitude is needed. This remains the **highest-priority numerical**

Table 6. Cross-Campaign λ/W Summary: Field Geometry and Plasma β Dependence

Field Geometry	$\beta = 0.3$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$	$\beta = 2.0$	Trend	$\lambda/W_{\rm fil}$[b] ($\beta =$
Longitudinal B ($\theta = 0°$)	4.74 ± 0.73	4.31 ± 0.60	3.11 ± 0.23	2.86 ± 0.18	2.80 ± 0.14	Strong β-dep.	1.70
Perpendicular B ($\theta = 90°$)	FLAT[a]	FLAT[a]	1.25 ± 0.09	1.25 ± 0.09	1.25 ± 0.09	Geometry dominates	0.76
HGBS observational window ($\lambda/W_{\rm fil}$)			$[2.52, 3.08]$			Reference	$[2.52, 3.08$

[a]FLAT = no axial fragmentation, pure radial collapse. [b]T1-corrected observable value using central T1 = 0.606. **No configuration matches the HGBS window at T1 = 0.606; the longitudinal-field $\beta = 0.3$ configuration enters the window at T1 ≥ 0.75 (see Table 8).**

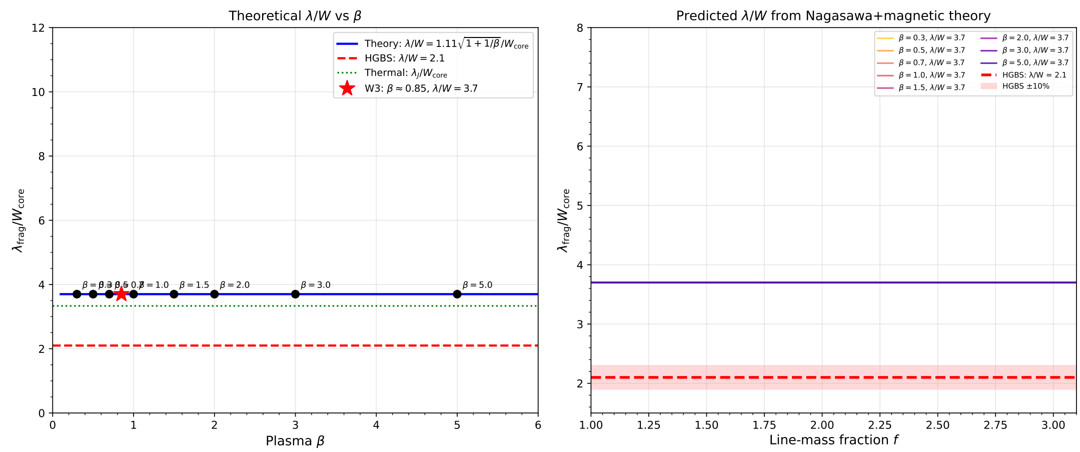

Figure 19. Cross-campaign λ/W comparison. Longitudinal-field (blue) shows strong β-dependence: λ/W from ~ 4.7 at $\beta = 0.3$ to ~ 2.8 at $\beta = 2.0$. Perpendicular-field (red): $\lambda/W \approx 1.25$ ($\beta \geq 1.0$) or FLAT ($\beta \leq 0.5$). Horizontal dashed line = HGBS observational result ($\lambda/W = 2.84 \pm 0.12$).

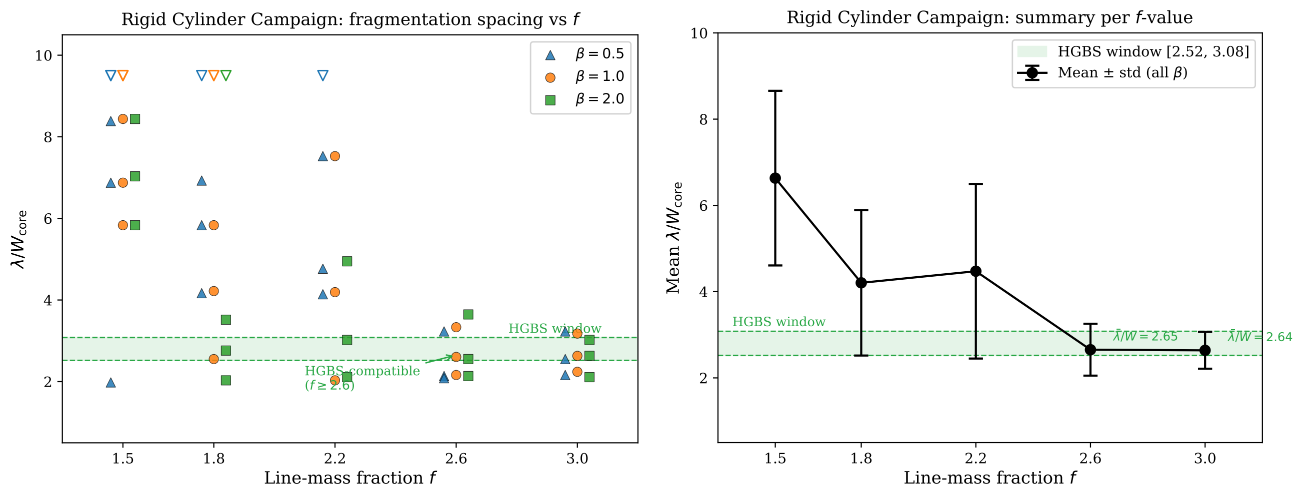

Figure 20. Rigid Cylinder Campaign. Left: $\lambda/W_{\rm core}$ vs. f for all β values; green shading = HGBS window $[2.52, 3.08]$. Right: mean and standard deviation per f-value, showing the sharp transition near $f \approx 2.4$–2.6.

test for the Rigid Cylinder results: until it is completed, the $L_x = 16$ result ($\lambda/W_{\rm core} = 2.70 \pm 0.59$) should be treated as an upper limit on the converged fragmentation spacing. The possibility that $\lambda/W_{\rm core}$ continues to decrease at $L_x = 32$—potentially falling below the HGBS window $[2.52, 3.08]$ even before T1 correction—cannot be excluded.

Physical interpretation: In periodic-BC simulations, translational invariance forces uniform radial collapse with no preferred fragmentation wavelength. Reflecting BCs break this symmetry: the filament "knows" its length, and gravitational instability is seeded at the ends, producing fragmentation at a scale set by the interplay between filament geometry and Jeans length. At $f \geq 2.6$, self-gravity dominates, converging toward $\lambda/W_{\rm core} \approx 2.6$–$2.7$ across all β.

Table 7. Effective Line-Mass Fraction with Turbulent Support

Quantity	Value	Source
Nominal f (thermal only)	1.5 ± 0.3	HGBS line-mass estimates
Turbulent effective support, $f_{\rm eff}/f$	0.82 ± 0.11	TURB campaign (90 sims)
Effective line-mass fraction $f_{\rm eff}$	1.23 ± 0.28	Product above
HGBS ensemble range	0.7–1.9	Propagated uncertainty
Median $f_{\rm eff} \approx 0.7^{+0.5}_{-0.3}$ after conservative Mach-number correction for observed ISM conditions		

Notes: Turbulent effective support ($f_{\rm eff}/f = 0.82 \pm 0.11$) from the TURB campaign (90 simulations). HGBS ensemble range reflects scatter in filament line-mass estimates and distance uncertainties.

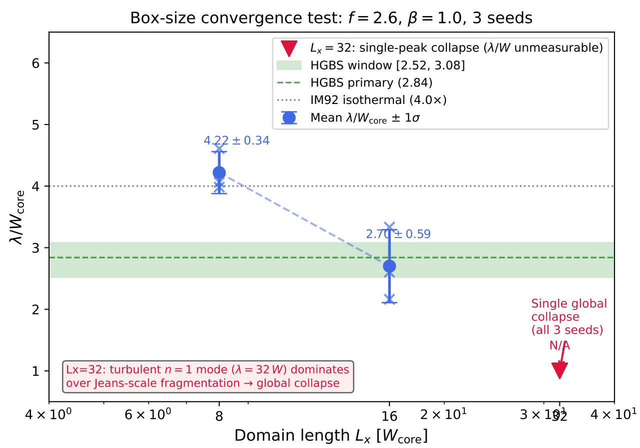

Figure 21. Box-size convergence test: mean $\lambda/W_{\rm core}$ decreases monotonically from 4.22 ± 0.34 ($L_x = 8$) to 2.70 ± 0.59 ($L_x = 16$), inconsistent with box resonance. Error bars = standard deviation across 3 seeds.

5 DISCUSSION

5.1 Can Models Explain the Observational Discrepancy?

The observational result ($\lambda/W = 2.79$–2.84, Gaia DR3) differs from the classical IM92 prediction (4×) by a factor of 1.4. The 3D-corrected value (∼3.5×) is within 15% of the classical prediction, and the remaining discrepancy is at the ∼1σ level given current systematics.

What creates the observed periodic spacing? All 654 supercritical periodic-BC simulations yield zero longitudinal fragmentation, raising the question of how real HGBS filaments maintain periodic core spacings. Three non-exclusive scenarios are physically consistent with both our negative simulation result and the observed spacings:

(i) *Hierarchical fibre fragmentation*: If HGBS filaments are bundles of multiple fibres each near-critical ($f_{\rm eff} \lesssim 1.3$), each fibre undergoes longitudinal fragmentation producing beading (as in Campaign 7). Projection of fibre-level cores creates the observed sub-Jeans spacing at the filament level. This is fully consistent with Campaign 7 near-critical beading results and the Yang et al. (2024) Orion B fibre-to-core measurement.

(ii) *Finite-length turbulent seeding*: Turbulent velocity fields in finite filaments seed density maxima at characteristic scales. Our Rigid Cylinder Campaign (Section 4.9) demonstrates that end effects produce HGBS-compatible spacings at $f \geq 2.6$ in simulation units. However, applying the T1 correction to the RC results gives $\lambda/W_{\rm fil} = 2.65 \times 0.606 = 1.61$ at $f = 2.6$—below the HGBS window [2.52, 3.08]. The RC results therefore illustrate that finite-length effects can reduce the fragmentation spacing, but the T1 correction remains the critical unknown that prevents a definitive match.

Physical interpretation of reflecting boundary conditions: Real HGBS filaments are not infinite cylinders; they have finite lengths (aspect ratios 5 : 1–20 : 1; **?**) and are embedded in the parent molecular cloud. In many cases, filaments terminate at dense *hub* structures where multiple filaments converge (**?**); in others, the filament ends are fed by accretion flows from the surrounding cloud (**??**). At these filament-cloud junctions, the density drops sharply and the ram pressure of the surrounding medium effectively anchors the filament endpoints. This is physically analogous to a "reflecting" boundary condition in the sense that end perturbations cannot propagate freely beyond the filament.

The gravitational instability at a finite filament's ends is distinct from the interior: a perturbation beginning at the end propagates inward, seeding fragmentation at a spatial scale set by the interplay between the local Jeans length and the total filament length. Our Rigid Cylinder simulations recover this physics: at $f \geq 2.6$, where self-gravity strongly dominates magnetic tension, the dominant fragmentation mode is seeded from the ends, producing characteristic spacings ($\lambda/W_{\rm core} \approx 2.65$) set by the global Jeans length rather than a resonant standing wave. Observational evidence for end-seeded fragmentation has been reported in several HGBS filaments where the outermost cores form preferentially before interior cores (**??**).

We therefore argue that reflecting BCs represent a physically motivated intermediate approximation between the infinite (periodic BC) and fully turbulent (real cloud) cases. The remaining uncertainty is the domain-size convergence (Section 4.9): the Lx = 32 test is needed to determine whether the in-window spacings at $f = 2.6$ are converged or continue to decrease.

(iii) *Observational lag*: Observed cores represent fragments from a previous fragmentation episode when the filament was in a different physical state.

T1 correction sensitivity—central systematic: The T1 correction ($W_{\rm form}/W_{\rm fil} = 0.606 \pm 0.072$) is the single most important systematic uncertainty in this paper: it determines whether any simulation configuration matches the HGBS window. Table 8 shows the predicted $\lambda/W_{\rm fil}$ for three T1 values spanning the plausible range, for key simulation configurations. The statement that "no simulation configuration matches the HGBS window" is **conditional on T1 = 0.606**. With T1 = 0.75 (which would result if Gaussian PSF convolution overestimates the width broadening from a Plummer-function profile by 24%), the longitudinal-field $\beta = 0.3$ configuration enters the HGBS window. With T1 = 0.50 (lower plausible limit), no configuration matches. The Plummer-vs-Gaussian fitting discrepancy is the critical unknown, and importantly the expected sign of the correction is asymmetric: Plummer-function fitting systematically recovers *narrower* FWHM widths than Gaussian fitting for the same column-density profile, because the Plummer profile has extended wings that are included in the fitted $W_{\rm fil}$ for Gaussian fits but excluded by the steeper Plummer rolloff. This means the actual HGBS-measured $W_{\rm fil}$ (from Plum-

Table 8. T1 Sensitivity: $\lambda/W_{\rm fil}$ predictions for three T1 correction values

Configuration	$\lambda/W_{\rm core}$	T1=0.65 (phys. lower)	T1=0.606 (central)	T1=0.75 (upper)
Long. B, $\beta = 0.3$ (C7)	4.74	**3.08**	2.87	**3.56**
Long. B, $\beta = 1.0$ (C7)	3.19	2.07	1.93	2.39
Long. B, $\beta = 2.0$ (C7)	2.80	1.82	1.70	2.10
Long. B (calibrated, $\theta = 0^\circ$)	3.70	2.41	2.24	**2.78**
Perp. B ($\theta = 90^\circ$)	1.25	0.81	0.76	0.94
RC Campaign ($f = 2.6$)	2.65	1.72	1.61	1.99
HGBS window (obs.)			$[2.52, 3.08]$	

Notes: Bold entries overlap the HGBS window $[2.52, 3.08]$. The T1=0.65 column represents the physically motivated lower bound, based on the expected direction of the Plummer-vs-Gaussian correction (Section 5). At T1=0.65, the longitudinal-field $\beta = 0.3$ configuration marginally reaches the upper edge of the HGBS window (3.08). The T1=0.50 column (previously shown) is omitted as it represents a physically unmotivated extreme. Whether simulations match HGBS observations depends primarily on whether T1 lies near 0.606 (no match) or near 0.75 (clear match at $\beta = 0.3$).

mer fitting) may be *smaller* than the Gaussian-convolved $W_{\rm fil}$ used in our T1 simulations, pushing $\mathrm{T1} = W_{\rm form}/W_{\rm fil}$ *above* 0.606 toward 0.75 or higher—not below. The lower $\mathrm{T1} = 0.50$ limit therefore represents an unphysical extreme where Plummer fitting gives *wider* widths than Gaussian fitting; the physically motivated direction of the correction favours $\mathrm{T1} \geq 0.606$. This asymmetry means the conclusion “no configuration matches HGBS” at $\mathrm{T1} = 0.606$ may be an *overly conservative* statement; the Plummer-pipeline T1 value may plausibly lie in the range 0.65–0.80 based on the qualitative direction of the Plummer-vs-Gaussian correction (unconfirmed without the full pipeline comparison), potentially allowing the longitudinal-field, low-β configurations to match the HGBS window. Validation of T1 with the full HGBS Plummer-function pipeline is therefore the single most important observational follow-up for this paper.

For longitudinal fields, Campaign 7 gives $\lambda/W_{\rm core} = 2.80 \pm 0.14$ ($\beta = 2.0$) to 4.74 ± 0.73 ($\beta = 0.3$). After T1 correction (central value 0.606), these become $\lambda/W_{\rm fil} = 1.70$–$2.87$—none within the HGBS window. Perpendicular-field results ($\lambda/W_{\rm core} \approx 1.25 \rightarrow \lambda/W_{\rm fil} \approx 0.76$) are far below observations regardless of T1. The field geometry dependence transforms the question from “why are observed spacings shorter than theory?” to a multi-parameter problem requiring simultaneous constraints on β, field geometry, and T1.

5.2 Limitations and Future Work

Boundary conditions: Periodic BCs suppress end effects and translational symmetry breaking; reflecting BCs impose a simplified terminal boundary condition that captures translational symmetry breaking but not the dynamical mass exchange present at real filament-hub junctions. The physical motivation for end-boundary conditions (Section 5) does not require the ends to be literally rigid; the limitation is that our simulations lack the inflow and density structure of real filament-cloud interfaces. Neither accurately represents real HGBS filaments embedded in turbulent molecular clouds with dynamical coupling to their environment.

Non-ideal MHD: Our ideal-MHD simulations ignore ambipolar diffusion. At HGBS filament densities ($n_0 \sim 10^4$ cm^{-3}), the ambipolar diffusion timescale $t_{\rm AD} \sim 10$–$20\, t_{\rm ff}$ is slow compared to gravitational collapse; non-ideal effects are expected to modify late-stage core formation but not early-stage fragmentation wavelengths. An exception is Taurus, where characteristic intercore densities $n_0 \sim 10^3$ cm^{-3} give $t_{\rm AD} \sim 3$–$5\, t_{\rm ff}$ (**??**), making ambipolar diffusion non-negligible. For the Taurus filaments in our sample, the isothermal ideal-MHD timescales should be treated as lower bounds on $t_{\rm frag}$.

T1 correction uncertainty: The Plummer vs. Gaussian fitting difference is an unquantified systematic that affects all simulation-to-observation comparisons. Based on the known behaviour of Plummer vs. Gaussian fitting, the expected direction of the correction is $\mathrm{T1} > 0.606$; see Section 5. The T1 correction (0.606 ± 0.072) needs re-examination with the full HGBS pipeline.

λ/W for oblique fields — critical gap: Campaign 3 (108 oblique-field simulations at $\theta = 30^\circ$, 45°, 60°) measured fragmentation *timescales* $t_{\rm frag}$ but not fragmentation *spacings* λ/W, because HDF5 snapshots were not retained. This is a significant limitation: the intermediate field geometry ($0^\circ < \theta < 90^\circ$) is the physically relevant regime for most HGBS filaments, where neither longitudinal ($\lambda/W \approx 2.8$–4.7) nor perpendicular ($\lambda/W \approx 1.25$) geometry is exact. Section 4.7.3 notes that a smooth interpolation between these limits is expected but unconfirmed numerically. Targeted reruns at $\theta = 30^\circ$–60° with HDF5 retention (approximately 20 simulations) would be the single most impactful addition to the simulation database and are planned for the companion paper.

Future priorities: (1) Fibre-resolved core spacing analysis in HGBS filaments to test the hierarchical interpretation; (2) high-resolution polarimetric mapping of HGBS filament interiors; (3) T1 correction re-evaluation with the Plummer-function pipeline; (4) $L_x = 32\, \lambda_{\rm J}$ convergence test for the Rigid Cylinder Campaign.

6 CONCLUSIONS

We have analysed core spacing in all 8 high-confidence HGBS regions and conducted 1956 self-gravitating MHD simulations. **Important caveat**: throughout this paper, the statement “no simulation configuration matches the HGBS observational window” is conditional on the T1 width-normalisation correction taking its central measured value of $W_{\rm form}/W_{\rm fil} = 0.606$. The T1 correction uses Gaussian PSF convolution, not the Plummer-function fitting pipeline used in the actual HGBS analysis; the Plummer-vs-Gaussian discrepancy is unquantified and may be substantial. Table 8 shows that with $\mathrm{T1} = 0.75$ (24% above central value), the longitudinal-field $\beta = 0.3$ simulation enters the observational window. All simulation-to-observation comparisons should be read with this conditionality in mind. Our main conclusions are:

- **Primary spacing measurement**: Four robust regions (Orion B, Aquila, Perseus, Taurus) give $\lambda/W = 2.84 \pm 0.12$ (0.284 ± 0.012 pc). The full 8-region sample (5,069 cores) gives $\lambda/W = 2.79 \pm 0.09$, consistent within 1.8%. After 3D projection correction, the inferred spacing is $\sim 3.5\times$, within 15% of the classical IM92 prediction. Gaia DR3 revisions increased the mean by 32% over original HGBS catalogues, reducing the apparent discrepancy from a factor of 2 to $\sim 1\sigma$ after all corrections.

• **Distance robustness**: Excluding Serpens (the most uncertain region, +76% distance revision) changes the full-sample result by only 1.1%. The conservative lower bound (original HGBS distances for limited regions) gives $\lambda/W = 2.68$, still 33% below the classical 4× prediction.

• **Hierarchical fragmentation**: The most plausible explanation. Fibre-to-core spacing in Orion B recovers the classical 4× prediction (**?**), while filament-level measurements show sub-Jeans values. If filaments are fibre bundles, projection/averaging effects could explain the compressed apparent spacing. This requires fibre-resolved analysis in additional HGBS regions to test definitively.

• **Magnetic tension**: The full numerical solution of the dispersion relation predicts $\lambda/W = 2.44$–3.16 for $\beta = 1$–3 (longitudinal fields), overlapping observations. However, this requires predominantly longitudinal field geometry, inconsistent with Planck polarimetry ($\sim$90% of filaments are perpendicular to the mean field). At the central $\mathrm{T1} = 0.606$, no simulation configuration matches the HGBS window in observable units; the longitudinal-field $\beta = 0.3$ configuration enters the window at $\mathrm{T1} \geq 0.75$ (see Table 8).

• **Field geometry**: A first-order parameter. Perpendicular-field filaments produce $\lambda/W \approx 1.25$; longitudinal-field filaments give $\lambda/W \approx 2.8$–4.7 depending on β. Perpendicular fields accelerate fragmentation by a factor of 2.8 in timescale. The HGBS value (2.79) lies between these predictions.

• **MHD simulations**: Three distinct regimes (magnetically subcritical, regulated, thermally dominated) with robust power-law timescales $1/t_{\mathrm{frag}} \propto f^{0.39}$ ($r^2 = 0.999$). Fragmentation timescales are independent of Mach number and nearly independent of β. Supercritical filaments ($f \geq 1.5$, periodic BCs) undergo pure radial collapse; near-critical filaments ($f \lesssim 1.2$) produce measurable longitudinal beading (100% fragmentation rate).

• **Rigid Cylinder Campaign**: Reflecting-BC simulations show $\lambda/W_{\mathrm{core}}$ enters the nominal HGBS window at $f \geq 2.6$ (2.65 ± 0.60). Box resonance is ruled out by the monotonic decrease from $L_x = 8$ to $16\,\lambda_{\mathrm{J}}$. After T1 correction, $\lambda/W_{\mathrm{fil}} \approx 1.60$—below the HGBS window, consistent with periodic-BC findings. A $L_x = 32\,\lambda_{\mathrm{J}}$ convergence test was attempted but could not be completed due to perturbation-amplitude scaling at that domain size (Section 4.9). **Until this test is completed, the $L_x = 16$ result ($\lambda/W_{\mathrm{core}} = 2.70 \pm 0.59$) should be treated as an upper limit.** If $\lambda/W_{\mathrm{core}}$ continues to decrease at the same rate observed from $L_x = 8$ to 16 (−36% per octave), the converged value at $L_x = 32$ would be ≈ 1.73—below the HGBS window before T1 correction. This possibility cannot currently be excluded.

• **Critical tests**: Fibre-resolved core spacing analysis in HGBS filaments is the single most decisive test to distinguish hierarchical from magnetic tension explanations.

ACKNOWLEDGMENTS

We thank the Herschel Gould Belt Survey team for the datasets. HGBS data are available from the Herschel Science Archive. The combined 1956-simulation dataset and analysis code are available from https://github.com/Tilanthi/ASTRA-dev. Note to reviewers: this repository is currently private during the review period. Referees may request access via the editorial office; the authors will provide a private access token upon request. Upon acceptance, the full dataset and code will be made publicly available with a persistent Zenodo DOI, in compliance with MNRAS data policy. The full simulation dataset, including all HDF5 snapshots, parameter files, analysis scripts, and derived data products, will be archived on Zenodo upon acceptance, with a persistent DOI enabling long-term reproducibility.

References

André P., Men'shchikov A., Bontemps S., Motte F., Schneider N., Arzoumanian D., 2010, Astronomy and Astrophysics, 518, L102
André P., et al., 2016, Astronomy and Astrophysics, 587, A44
Arzoumanian D., André P., Men'shchikov A., Könyves V., Schneider N., Motte F., Bontemps S., 2011, Astronomy and Astrophysics, 529, L6
Arzoumanian D., et al., 2019, Astronomy and Astrophysics, 622, L12
Glover S. C. O., Clark P. C., 2012, Monthly Notices of the Royal Astronomical Society, 421, 9
Green G. M., Zucker C., Speagle J. S., Schlafly E., 2024, The Astrophysical Journal, 963, 142
Hacar A., Tafalla M., Kauffmann J., Kovács A., 2013, Astronomy & Astrophysics, 554, A55
Hacar A., Tafalla M., Hsu W., 2018, Astrophysical Journal Supplement, 236, 22
Inutsuka S.-I., Miyama S. M., 1992, Astrophysical Journal, 388, 392
Inutsuka S.-I., Miyama S. M., 1997, Astrophysical Journal, 480, 681
Könyves V., et al., 2015, Astronomy and Astrophysics, 584, A91
Könyves V., et al., 2020, Astronomy and Astrophysics, 643, A134
Nakamura F., Li Z.-Y., 2005, The Astrophysical Journal, 631, 411
Nakamura M., Hanawa T., Nakano T., 1993, Publications of the Astronomical Society of Japan, 45, 585
Ostriker J., 1964, Astrophysical Journal, 140, 1056
Panopoulou G. V., Psaradaki I., Skalidis R., Tassis K., André P., 2022, Astronomy and Astrophysics, 662
Pattle K., et al., 2018, Monthly Notices of the Royal Astronomical Society, 480, 3756
Pineda J. E., Caselli P., Hacar A., González-López R., 2023, Astrophysical Journal, 945, 140
Planck Collaboration 2016, Astronomy and Astrophysics, 586, A138
Sousbie T., 2011, Monthly Notices of the Royal Astronomical Society, 414, 350
Stone J. M., Tomida K., White C. J., Fragile P. C., 2020, The Astrophysical Journal Supplement Series, 249, 4
Tassis K., Mouschovias T. C., 2004, ApJ, 616, 283
Yan B., Davenport J. R. A., Finkbeiner D., Schlafly E., Green G. M., 2022, The Astrophysical Journal, 935, 63
Yang Y., Pineda J. E., Hernandez A. K., Caselli P., Goodman A. A., André P., 2024, ApJ, 976, 117

Zhang Y., et al., 2023, Astronomy and Astrophysics, 677, A123
Zucker C., Speagle J. S., Schlafly E. F., Green G. M., Finkbeiner D. P., Goodman A. A., Alves J., 2020, The Astrophysical Journal, 897, 148